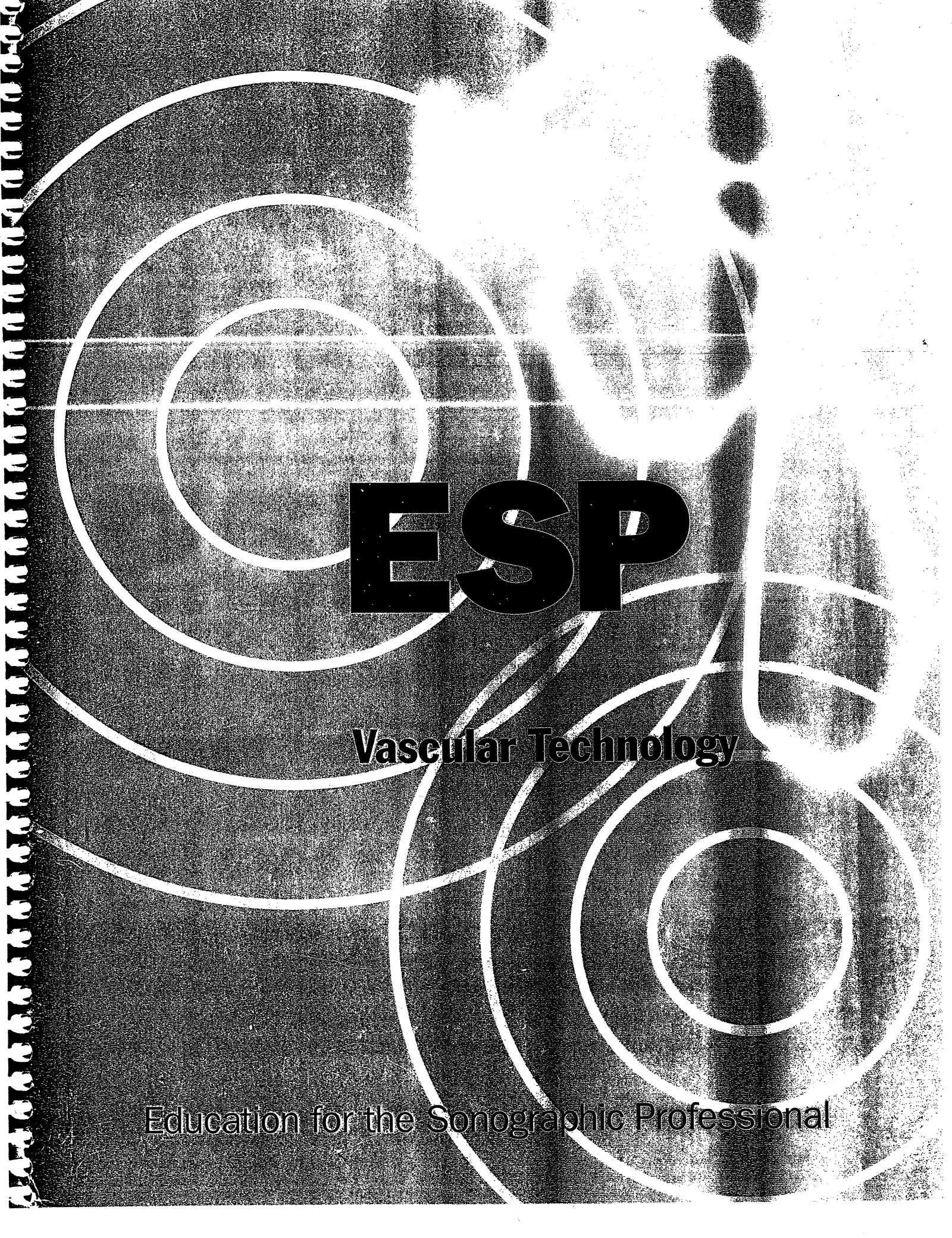




ESP

Vascular Technology

Education for the Sonographic Professional

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This course is intended to provide the practicing vascular technologist/sonographer with an intensive review for certification. This syllabus adheres to the ARDMS study outline content, as well as to the CCI study outline; and should be used in conjunction with other pertinent references and resources. Although designed to assist the experienced practitioner in the successful completion of the certification examination, this course also provides those with less experience, a strong educational foundation upon which to build.

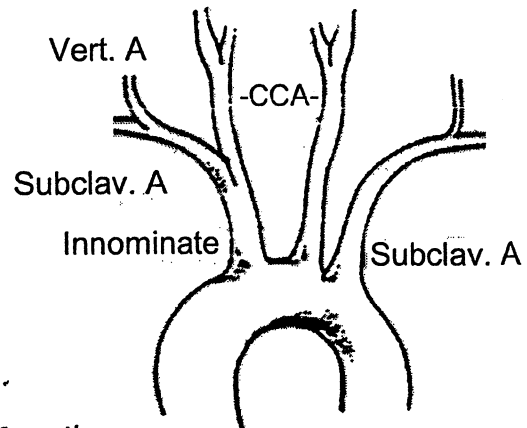
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A. AORTIC ARCH

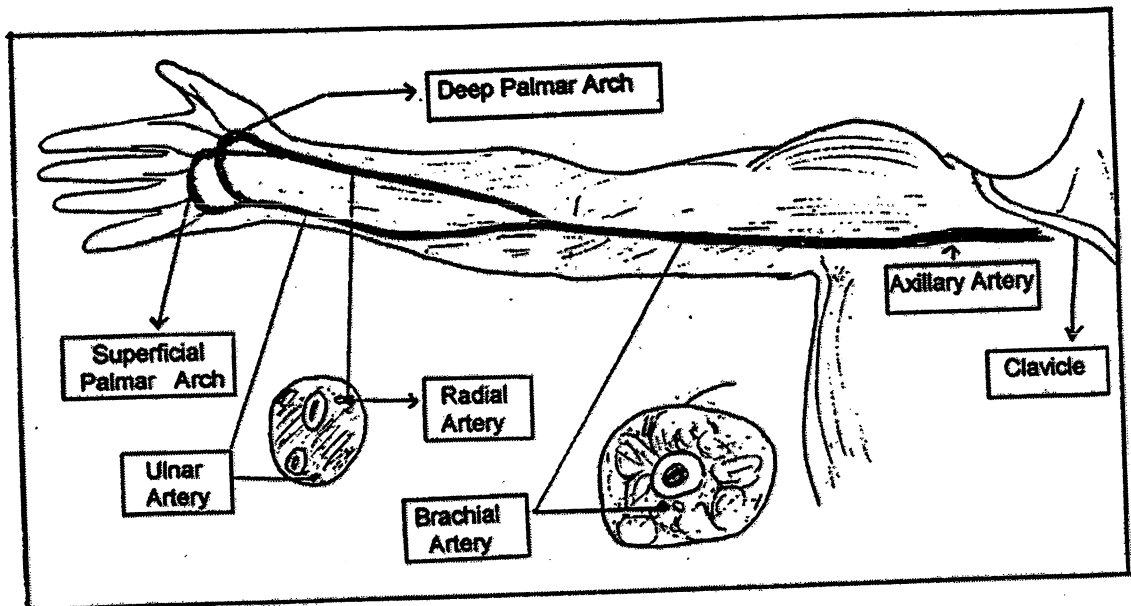
1. Proximal to Arch, the first branches of ascending aorta are the coronary A.
2. Innominate / Brachiocephalic
 - Divides into the right common carotid and subclavian arteries
3. Left common carotid artery (LCCA)
4. Left subclavian artery



B. UPPER EXTREMITY ARTERIES

1. Subclavian artery:
 - Runs laterally to outer border of 1st rib becoming axillary A.
 - Some branches: vertebral, thyrocervical, costocervical.
2. Axillary artery: after giving off several branches, becomes brachial A.
3. Brachial Artery: Branches into the radial and ulnar arteries at the inner aspect of the elbow, (also known as the antecubital fossa).
4. Radial artery: travels down lateral side of forearm into hand, branching to form:
 - Superficial palmar (volar) arch;
 - Terminates in deep palmar arch by joining deep branch of ulnar artery.
5. Ulnar artery: travels down medial side of forearm into hand, branching to form:
 - Deep palmar (volar) branch;
 - Terminates in superficial palmar arch.
 - Predominate source of blood flow to the hand
6. Superficial palmar (volar) arch includes:
 - Distal portion of the ulnar artery
 - Branch of the radial artery.

7. Deep palmar arch includes:
 - Deep palmar branch of ulnar artery
 - Distal portion of the radial artery.
8. Digital arteries arise from the palmar arches and extend into the fingers dividing into lateral and medial branches.



C. ABDOMINAL AORTA

1. Visceral branches

a. Celiac artery: (CA)

- Supplies stomach, liver, pancreas, duodenum, spleen.
- Branches into L. gastric, splenic, and common hepatic arteries.

b. Superior mesenteric artery: (SMA)

- Supplies the small intestine, cecum, parts of colon.
- Located about 1 cm behind the celiac A.
- Can be a common trunk of the celiac A. and SMA.

c. Renal arteries: Find L. renal vein to find artery

- Supply blood to the kidneys, suprarenal glands, ureters.
- Multiple renal arteries not uncommon bilaterally.

- In transverse, a landmark for locating left renal A, is the left renal V which crosses the aorta anteriorly; the artery being just posterior.

d. Inferior mesenteric artery: (IMA)

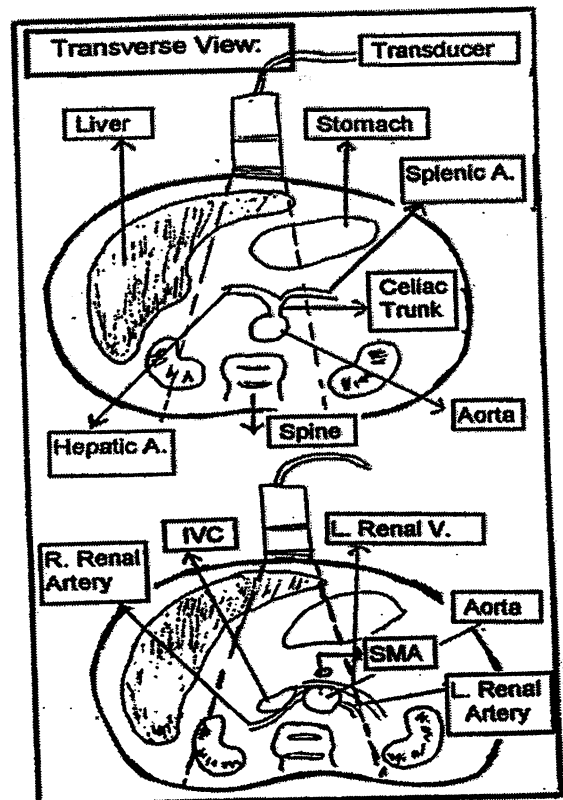
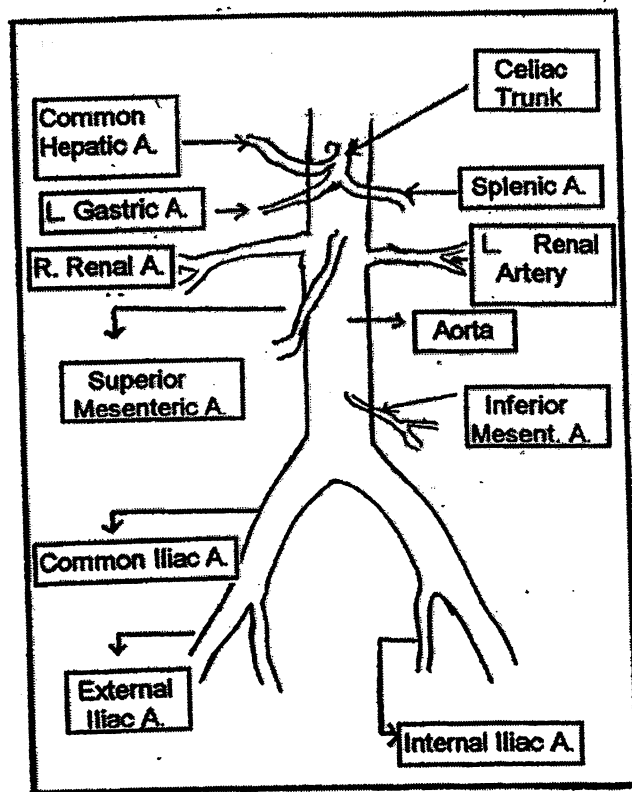
- Supplies transverse, descending colon and part of rectum.
- Arises from abdominal aorta about 3-4 cm above bifurcation
- Can act as a collateral connection

2. Terminal branches

- a. Both common iliac arteries (CIA) are the most distal branches of the aorta carrying blood to the pelvis, abdominal wall and lower limbs. The CIA divides into:

- Internal iliac artery (hypogastric).
- External iliac artery: travels along medial side of Psoas major muscle.

Passing underneath inguinal ligament, becomes the CFA.



D. LOWER EXTREMITY ARTERIES

1. Common femoral artery (CFA) divides into:

*After Hunter's canal
CFA becomes SFA*

a. Superficial femoral artery (SFA)

- Runs the length of the thigh, passing through an opening in the tendon of the Adductor hiatus (adductor canal, or Hunter's canal),
- Enters the popliteal fossa behind the knee.

b. Deep femoral (profunda femoris) artery

- A large branch; arises about 5 cm. from the inguinal ligament on the lateral side.
- Can act as a collateral connection.

2. Popliteal artery

- Adductor hiatus: termination of the SFA and beginning of popliteal artery.
- Gives off number of genicular branches (can act as collaterals) to supply muscles, knee joint and skin.
- At interval between tibia and fibula, divides into anterior and posterior tibial arteries.

3. Anterior tibial (ATA)

- First branch off distal popliteal artery.
- At its lower end, becomes dorsalis pedis artery (DPA) and is directed across dorsum of foot towards base of great toe.

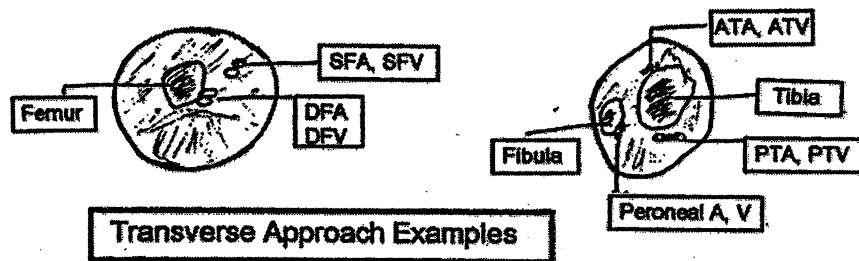
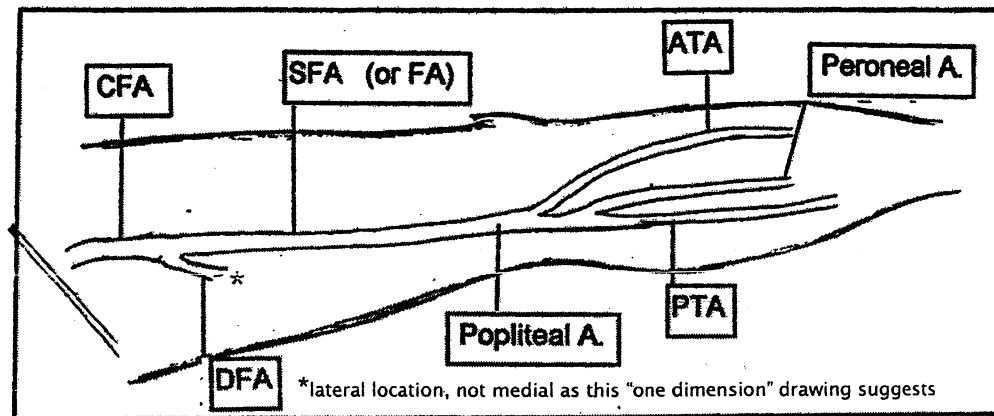
Major branch of DPA: deep plantar artery; penetrating the sole of foot, it unites with lateral plantar artery to complete plantar arch

4. Posterior tibial (PTA)

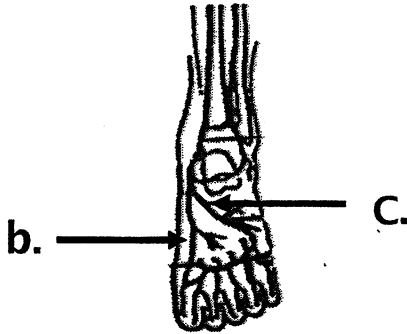
- Extends obliquely down posterior/medial side of the leg.
- Tibioperoneal trunk: short segment between ATA branch and branches of PTA and peroneal arteries.
- Major branches: lateral and medial plantar arteries, branching below medial malleolus to supply sole of foot.

5. Peroneal artery

- a. Passes toward fibula, traveling down medial side of that bone to supply structures of the lateral side of leg / foot.



6. Digital arteries/Plantar arch



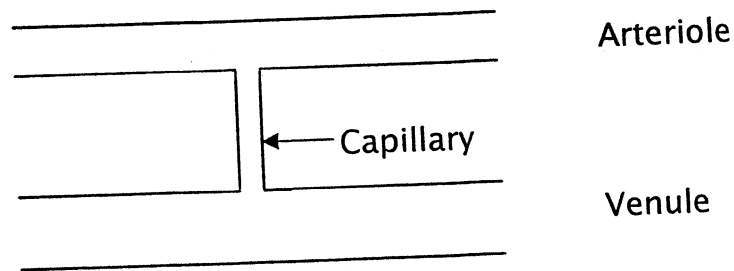
a. The plantar and dorsal metatarsals distribute blood to digits.

b. The plantar arch consists of deep plantar artery (branch of DPA) and

c. The lateral plantar artery (branch of PTA) which unites with the deep plantar artery.

7. Vessels of the microcirculation:

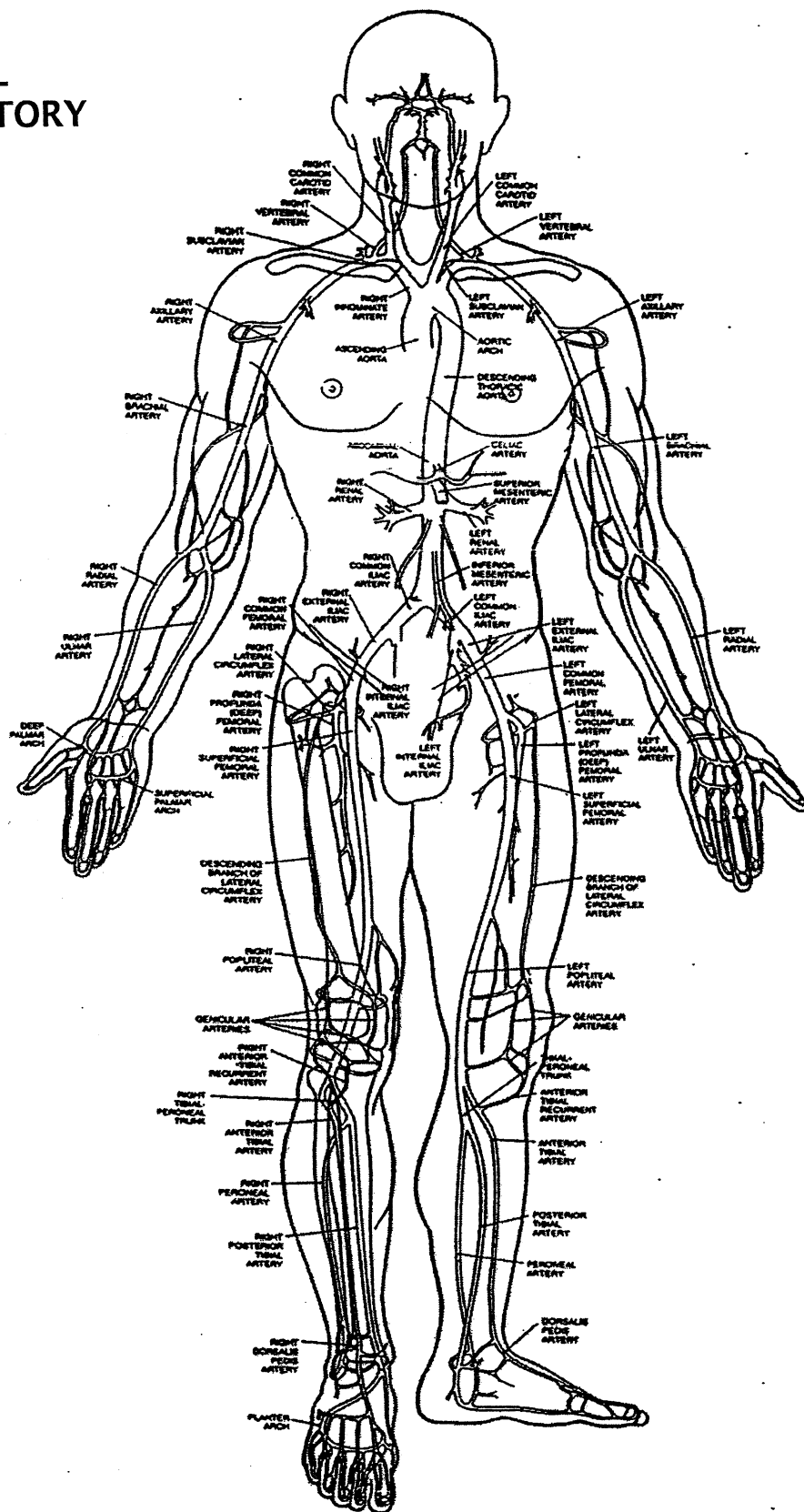
- a. Arteries transport gases, nutrients and other essential substances to the capillaries. Arteries progressively decrease in size from aorta (largest) to arterioles (smallest).
- b. Arterioles: considered resistance vessels; assist with regulating blood flow through contraction and relaxation.
- c. Capillaries: nutrients and waste products are exchanged between the tissue and blood.



E. **MICROSCOPIC ANATOMY OF THE ARTERIAL WALL**

1. Tunica intima/inner layer is thin, consisting of a surface layer of smooth endothelium, base membrane and connective tissue.
2. Tunica media/intermediate layer is thicker, composed of smooth muscle and connective tissue, largely of the elastic type.
3. Tunica externa/outer layer (adventitia) is somewhat thinner than media, contains fibrous connective tissue; some muscle fibers.
4. As a rule, the adventitial layer contains the vasa vasorum, tiny vessels that carry blood to the walls of the larger arteries.

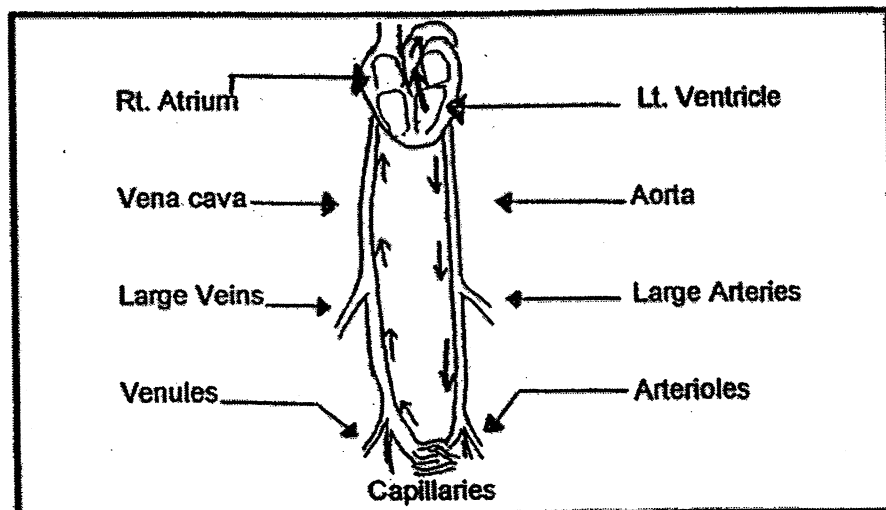
ARTERIAL CIRCULATORY SYSTEM



SOURCE:
Medasonics,
Mountain View, CA

A. THE ARTERIAL SYSTEM

1. Multi-branched elastic conduit set into oscillation by each beat of the heart.
2. Each beat pumps about 70 milliliters of blood into the aorta causing a blood pressure pulse.
3. Cardiac contraction begins:
 - Pressure in the left ventricle rises rapidly
 - Left ventricle pressure exceeds that in the aorta
 - Aortic valve opens, blood is ejected, BP rises
4. Increased heart rate delivers an increased blood volume.
5. Patient's cardiac status plays an important role in the movement of blood throughout the vascular system.
6. The heart pump:
 - Generates the pressure to move the blood.
 - Results in a pressure wave (energy wave) that travels rapidly throughout the system, demonstrating a gradual transformation as it travels distally.

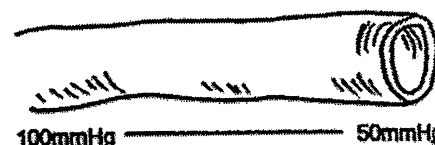


Example: Averaged sized supine patient:

- Arteries and veins are nearly the same level as the heart.
- There is 0 mmHg (HP) against the arteries and veins at the ankle. (Ankle P = circulatory P plus about 0 mmHg)
- When standing, HP increases, adding about 100 mmHg against ankle vessels. (Ankle P = circulatory P plus 100mmHg)

4. An **energy gradient** is needed to move blood from one point to another. The greater the gradient; greater the flow.

- Inertia:** relates to the tendency of a fluid to resist changes in its velocity (i.e. body at rest tends to stay at rest).
- As the blood moves farther out to the periphery, energy is dissipated largely in the form of heat.
- Energy continually restored by pumping action of heart.



C. **FACTORS AFFECTING RESISTANCE TO FLOW**

1. Movement of a fluid (blood) is dependent upon: physical properties of the fluid and what its moving through.

2.

$$R = 8\eta L / \pi r^4$$

R = Resistance η = viscosity L = vessel length
r = radius

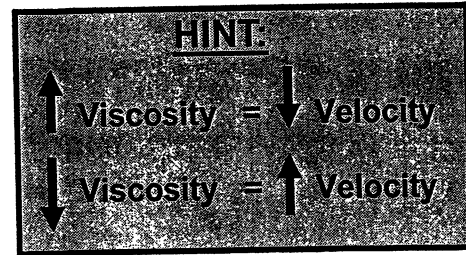
- "R" is directly proportional to variables in numerator: (viscosity and length)
- "R" is inversely proportional to the variable in the denominator: (radius)

3. Although **viscosity** and vessel length have an effect on **resistance**, a change in vessel diameter has a more dramatic effect.

4. **Internal friction** within a fluid is measured by its **viscosity**.

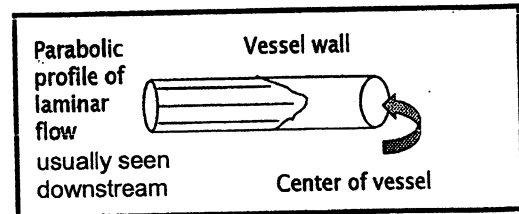
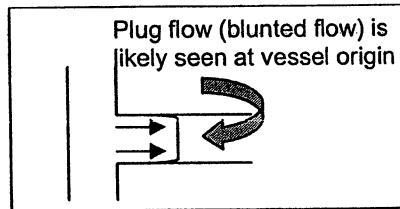
- Energy lost in form of heat (layers of rbc's rub against each other).
- Elevated hematocrit increases blood **viscosity** (thickness of a fluid); *increase resistance*
while severe anemia decreases blood **viscosity**. — *decrease resistance*

- c. Diminishing vessel size increases frictional forces and heat energy losses.



D. LAMINAR FLOW

1. Consists of layers of fluid particles moving against one another.
2. The fastest moving flow is in center; stationary layer remains at the wall. Laminar flow is considered stable flow.



E. ENERGY LOSSES

1. **Viscous** energy loss is due to increased friction between molecules and layers which ultimately causes energy loss.
2. **Inertial** losses occur with deviations from laminar flow, due to changes in direction and/or velocity.
 - The parabolic flow profile becomes flattened
 - Flow moves in a disorganized fashion.
 - This type of energy loss occurs at the **exit** of a stenosis.

F. POISEUILLE'S EQUATION

1. Defines relationship between: pressure, volume flow, resistance.
2. Helps answer question of how much fluid moves through vessel.

$Q = P / R$

Q = Volume Flow P = Pressure
 R = Resistance

Remember this comparison:

Hemodynamics	Electricity (Ohm's Law)
$Q = P / R$	$I = V / R$ ($I = E / R$)
Flow (Q)	= Current (I)
Pressure (P)	= Voltage (V) or (E)
Resistance (R)	= Resistance (R)

3. **Poiseuille's equation** and the relationship between the contributors to flow and the flow itself is illustrated:

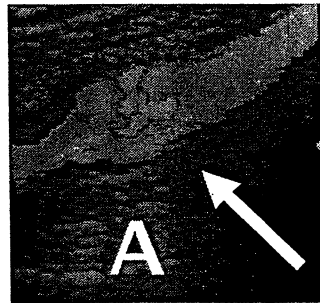
$$Q = \frac{(P_1 - P_2) \pi r^4}{8 \eta L}$$

Q = Volume Flow
 P₁ - P₂ = Pressures at proximal/distal ends
 r = Radius of the tube
 L = Length of the tube
 η = Viscosity of the fluid

- Diameter change has most dramatic effect on resistance.
- Radius (r⁴) of vessel is directly proportional to volume flow.
- Small changes in radius may result in large changes in volume flow.

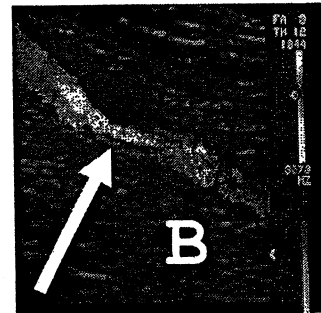
- G. **VELOCITY CHANGES** : The Law of Conservation of Mass explains the relationship between velocity and area. ($Q = A \times V$)

area ↓ - V ↑
area ↑ - V ↓



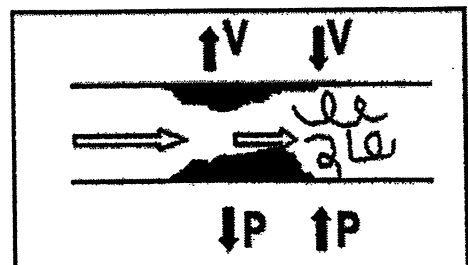
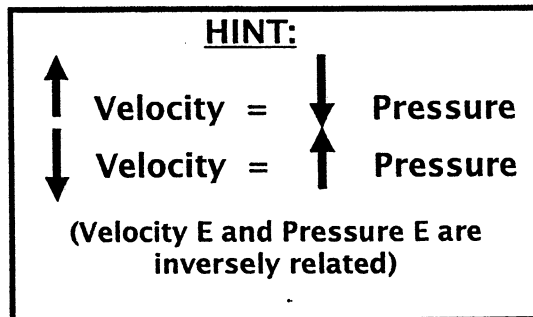
$$A = \downarrow V$$

$$B = \uparrow V$$

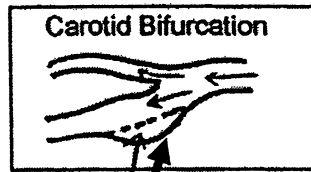


H. **PRESSURE / VELOCITY RELATIONSHIPS—Bernoulli**

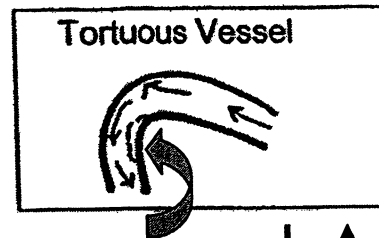
- Total energy contained in moving fluid is the sum of pressure, kinetic and gravitational energies.
- If one changes, the others make up for difference in order to maintain the original total fluid energy amount. (Performing exam on supine patient, there is NO change in hydrostatic pressure).



3. Pressure gradients (flow separations) are set up. They occur because of a geometry change with or without intra-luminal disease; and because of curves.

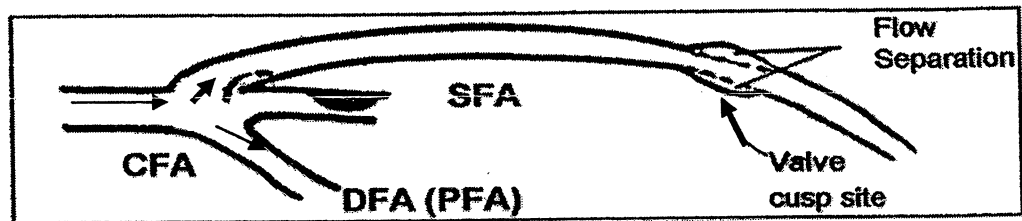


Flow separation: $V \downarrow P \uparrow$



Flow Separation: $V \downarrow P \uparrow$

4. Flow separations result in regions with stagnant or little movement. Example: bypass graft anastomosis site, or valve cusp site.



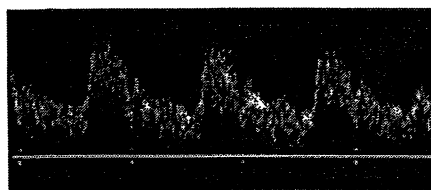
I. STEADY FLOW vs PULSATILE FLOW

1. Steady flow originates from a steady driving pressure.
 - a. Easy to deal with because behavior is more predictable.
 - b. In a rigid tube, energy losses are mainly **viscous**; can be described by **Poiseuille's equation**.
2. Pulsatile flow changes both the driving pressure conditions as well as the response of the system.
 - a. Systole: forward flow throughout the periphery. (Fluid Acceleration)
 - b. Late systole/Early diastole: temporary flow reversal, due to a phase shifted negative pressure gradient and peripheral resistance, causing reflection of the wave proximally.
 - c. Late diastole: flow is forward again, as reflective wave hits the proximal resistance of the next oncoming wave, and reverses.

J. PERIPHERAL RESISTANCE

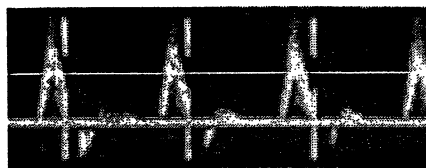
1. Low resistance flow: flow of a continuous (steady) nature feeding a dilated vascular bed.

Example arteries: ICA, vertebral, renal, celiac, splenic, hepatic.



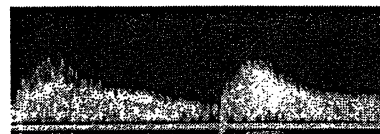
2. High resistance flow: flow of a pulsatile nature. Between incident pulses, hydraulic reflections travel back up the vessel from the periphery producing flow reversals in the vascular compartment.

Example arteries: ECA, subclavian, aorta, iliac, extremity arteries, fasting SMA.



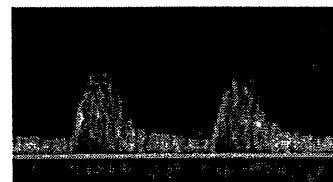
3. The reversal component of a high resistant signal may disappear distal to a stenosis because of decreased **peripheral resistance**, secondary to ischemia.

Doppler flow distal to a significant stenosis is lower resistant. In addition it's more rounded in appearance and is weaker in strength.



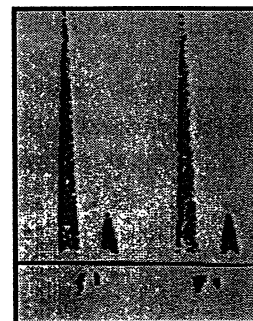
4. A normally high resistant (biphasic or triphasic) signal may become monophasic as it approaches the significant stenosis and/or arterial obstruction.

Doppler flow proximal to a significant stenosis is higher resistant in quality (could have no diastole or minimal diastole (as seen in this example)).



5. VASOCONSTRICTION

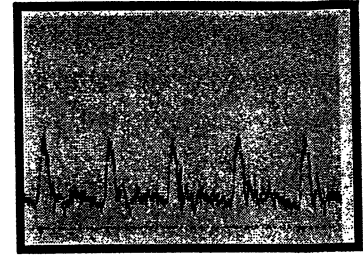
Pulsatile changes in medium/small sized arteries of the limbs are increased. When this occurs, pulsatility changes are usually decreased in the minute arteries.



distal flow is going to be lower

6. VASODILATATION

Pulsatile changes in medium/small sized arteries of the limbs are decreased (lower resistant). When this occurs, pulsatility changes are increased in minute arteries,

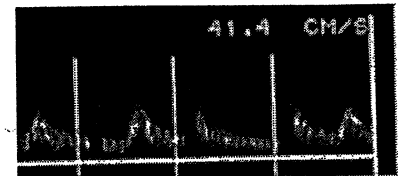


NOTE: As the inflow pressure falls as a result of stenosis; the natural response in periphery is to vasodilate to maintain flow.

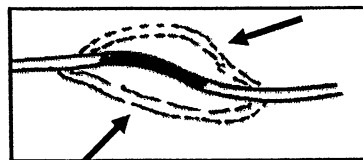
K. COLLATERAL EFFECTS

1. At rest, total blood flow may be fairly normal even in the presence of stenosis/ complete **occlusion** of main artery. Why? Development of a **collateral** network, and a compensatory decrease in **peripheral resistance**.
2. Arterial obstruction may alter flow in collateral channels nearby or further away from site of obstruction. Changes include:

- increased volume flow
- reversed flow direction
- increased velocity
- waveform pulsatility changes

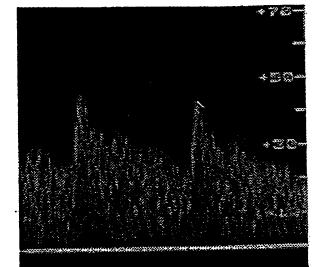


3. Location of collaterals helps provide tentative location of the obstruction.



L. EFFECTS OF EXERCISE

1. Exercise should induce peripheral vasodilatation which lowers the distal peripheral resistance, increasing blood flow.
2. Vasoconstriction and vasodilatation of blood vessels within skeletal muscles also influenced by sympathetic innervation fibers functioning primarily for regulation of body temperature.
3. **Exercise** is probably best single vasodilator of resistance vessels within skeletal muscle.



4. **Autoregulation:** ability of most vascular beds to maintain constant level of blood flow over a wide range of perfusion pressures.

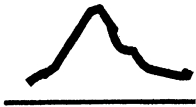
- Not present: perfusion pressure drops below a critical level.

BP rise = constriction of resistance vessels

BP fall = dilatation of resistance vessels

5. By decreasing resistance in the working muscle, **exercise** usually decreases "reflection" in exercising extremity.

- Example: a low resistant, monophasic flow signal is normally present in extremity arteries after vigorous exercise. The exercise causes reduced flow resistance (vasodilatation).



Same monophasic pattern also seen pathologically. Peripheral dilatation occurs in response to proximal arterial obstruction.

6. Higher resistant signals may occur from normal (physiologic) vasoconstriction at the arteriolar level OR from distal arterial obstruction.



M. **ADDITIONAL NOTES:**

- Flow to a cool extremity (vasoconstriction) will have pulsatile signals.

- Flow to a warm extremity (vasodilatation) will have continuous, steady signals.

- Pulsatility changes do not differentiate well between occlusion and severe stenosis.

- Waveforms may not be altered with good collateralization.

Distal effects of obstructive disease may only be detectable following stress (i.e., exercise or hyperemic evaluation).

N. **EFFECTS OF STENOSIS ON FLOW CHARACTERISTICS**

1. Review: Laminar flow

- Even distribution of frequencies at systole: lower frequencies distributed at the walls, (boundary layer); faster moving flow in centerstream.

2. A hemodynamically significant stenosis causes a notable reduction in volume flow and pressure.

Cross sectional area reduction of 75% = diameter reduction of 50%.

3. Effects of flow abnormality produced by a stenosis depends on factors such as:

- a. Length, diameter, shape, degree of narrowing
- b. Multiple obstructions in the same vessel: resistance to flow is additive; it results in a higher resistance than in each individual narrowing.
- c. Obstructions in different vessels that are parallel: resistance to flow is less than the resistance in each individual narrowing because only part of the blood flow is going through each narrowing.
- d. Pressure gradient; peripheral resistance beyond stenosis

4. STENOSIS PROFILE

a. Proximal to a stenosis: flow frequencies are usually dampened, with or without disturbance.

b. Stenosis:

Entrance into the stenosis produces an increase in Doppler shift frequencies (DSF), resulting in spectral broadening and **elevated velocities**.

- **Flow disturbance** occurs due to interrupted flow stability with high velocities and eddy currents.
- Abnormal "jet" (elevated velocities) may be isolated to area of stenosis, but also approaching and/or leaving it.

c. Post-stenotic turbulence:

A



B



C



Spectral broadening Vertical

- At stenosis exit, flow reversals, flow separations, vortices /eddy currents occur near edge of flow pattern.
- Flow quality is comprised of multiple changes in **direction** and spectral broadening as displayed by the spectral pattern.
- Energy expended as "heat" as eddys and vortices work against blood viscosity.



A B C
Pre-Stenosis Stenosis Post-stenosis

Note: After patient referred for pertinent medical necessity, important to obtain complete history, provide limited physical exam, and clinical information necessary to relate to test findings.

Essential to maintain a warm environment for the patient in order to permit warming (and peripheral dilation) to occur. An adequate amount of acoustic coupling gel is applied to the skin for all types of Doppler exams.

A. SIGNS AND SYMPTOMS

1. Chronic occlusive disease

a. Claudication:

- Pain in muscles usually occurring during exercise; subsides with rest.
- Results from inadequate blood supply to muscle.
- Discomfort is predictable and subsides within minutes after exercise.
- Level of disease usually proximal to location of symptoms.
- Pseudo-claudication mimics vascular symptoms but is neurogenic or orthopedic in origin

b. Ischemic rest pain:

- A more severe symptom of diminished blood flow.
- Occurs when limb not dependent; BP decreased (such as when sleeping).

c. Tissue loss:

- Necrosis or death of tissue.
- Due to deficient or absent blood supply.

NOTE: A patient history may state: 4 block claudication. This means the patient c/o pain after walking 4 blocks.

2. Acute arterial occlusion

- a. Symptoms include the 6 P's: pain, pallor, pulselessness, paresthesia, paralysis, polar.
- b. May result from thrombus, embolism or trauma.
- c. Emergency situation since the abrupt onset does not provide for the development of collateral channels.

3. Vasospastic Disorders

- a. Raynaud's phenomenon: a condition that exists when symptoms of intermittent digital ischemia occur in response to cold exposure or emotional stress.
- b. Changes in skin color may include pallor (whiteness), cyanosis (bluish color), or rubor (dark red color).
- c. PRIMARY RAYNAUD'S
 - Ischemia due to digital arterial spasm.
 - Common in young women; may be hereditary, bilateral; history of symptoms for 2 years without progression/evidence of cause.
 - Benign condition with excellent prognosis.
- d. SECONDARY RAYNAUD'S
 - Also known as obstructive Raynaud's syndrome.
 - Normal vasoconstrictive responses of arterioles superimposed on a fixed artery obstruction. Ischemia constantly present.
 - May be the first manifestation of Buerger's disease.

4. Physical Exam

Skin Changes

a. Color:

PALLOR: Result of deficient blood supply; skin pale.

RUBOR: Suggests dilated vessels or vessels dilated secondary to reactive hyperemia; skin is reddened.

CYANOSIS: A concentration of deoxygenated hemoglobin, causes bluish discoloration.

b. Temperature

- Touch patient's skin to determine warm/cold.
- Utilize skin thermometer to document precisely.

c. Lesions (Refer to Venous Section - Comparison chart)

- Ulcerations located: tibial area, foot, toes.
- Deep and more regular in shape.
- Quite painful as compared to venous ulcers.
- Duration of ulcer(s) is important, (i.e., days, weeks)
- Gangrene: death of tissue; usually due to deficient or absent blood supply.

d. Capillary filling

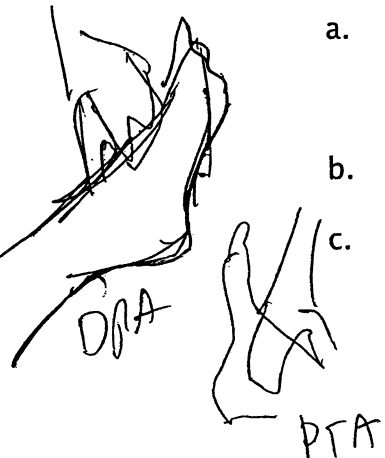
- An increase in the capillary refill time denotes decreased arterial perfusion.

e. Elevation/dependency changes

- Cadaveric pallor during elevation with ruborous red discoloration with dependency.

Palpation (pulses, aneurysms)

- a. Rhythmic throbbing of artery in time with heartbeat signifies adequate circulatory status. Diminished/absent pulse suggests arterial insufficiency.
- b. Grading pulses on scale of 0 (none) - 4+ (bounding) is fairly standard.
- c. Aneurysms can be palpated and described as bounding.



- d. Palpable "vibration" or "thrill" over pulse site may indicate a fistula, post-stenotic turbulence, or a patent dialysis access site.

- e. **PALPABLE PULSES:** aorta, femoral, popliteal, dorsalis pedis (DPA), posterior tibial (PTA)
- The peroneal artery is not palpable

Auscultation (bruits)

- Bruit auscultation is more often done with carotid examination.
- Other than stethoscope use for carotid vessels and heart, additional sites include: aorta, femoral, and popliteal arteries.
- Additional detail about bruits is located in the Cerebrovascular Testing section, Chapter 17

B. RISK FACTORS

1. Diabetes

- a. Atherosclerosis: more common; occurs at a younger age.
- b. Higher incidence of disease: distal pop. and tibial arteries.
- c. Medial calcification develops in LE arteries.
- d. Poor sensation (neuropathy) may lead to increased injury
- e. Higher incidence of gangrenous change, amputations.

2. Hypertension

- a. Unclear whether high blood pressure is a causative factor or enhances the development of the atherosclerotic process.
- b. Systemic hypertension is associated with greater incidence of coronary atherosclerosis. Increased BP taxes heart.

3. Hyperlipidemia

- a. Elevated plasma lipids closely assoc. with development of atherosclerosis.
- b. Frequent cause: Diet high in animal fat; metabolic problems associated with heredity.

4. Smoking

- a. Studies suggest the chemicals in cigarettes irritate the endothelial lining of the vessels, causing vasoconstriction.

5. Other (not controllable): age, family history, male gender

C. **MECHANISMS OF DISEASE** (Applicable to all arterial systems)

1. **Atherosclerosis (obliterans)**

- a. Most common arterial pathology: thickening, hardening, loss of elasticity of the artery walls.
- b. Changes occur in intima and media layer of the vessel.
- c. Major risk factors: smoking, hyperlipidemia, family history. Less important factors: hypertension, diabetes, sedentary lifestyle and arterial wall shear/stress.
- d. Most common sites:
 - Carotid bifurcation
 - Vessel origins
 - Infra-renal aorto-iliac system
 - CFA bifurcation
 - SFA at the adductor canal level
 - Trifurcation region
- e. Example: Leriche Syndrome—caused by obstruction of the aorta; usually occurs in males and is characterized by:
 - Fatigue in hips, thighs, or calves with exercise
 - Absence of femoral pulses
 - Impotence
 - Often times, pallor and coldness of LE

2. **Embolism**

- a. Obstruction of vessel by foreign substance or blood clot.
- b. Emboli may be solid, liquid or gaseous; may arise from the body or enter from without.
- c. Most frequent cause: small plaque breaks loose (e.g. atherosclerotic lesion, arteritis, or angiographic procedure) and travels distally until it lodges in small vessel.

Example: Blue toe syndrome. Toe ischemia results; often improves, mainly from other smaller branches.

3. Aneurysm

- a. True aneurysm is dilatation of all three arterial wall layers
Examples:

Fusiform: diffuse, circumferential dilatation.

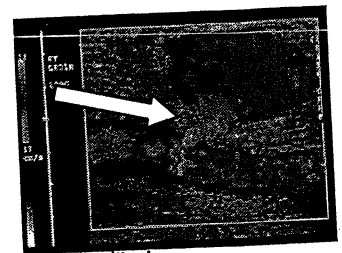


Saccular: localized out-pouching



- b. Dissecting aneurysm occurs when a small tear of the inner wall allows blood to form cavity between two wall layers. Often occurs in thoracic aorta.

- c. Pseudoaneurysm results from a defect in main artery wall (e.g., post catheter insertion). Must be a channel communication from main artery to pulsatile structure outside vessel walls.



- d. Most common location of a true aneurysm is infrarenal aorta. Other locations include: thoracic aorta, femoral, popliteal, renal.

- Patients with one aneurysm have higher incidence of 2.
- Cause unknown; ? poor nutrition, congenital defect, infection, or atherosclerosis.

Most frequent complication: Rupture of the aortic aneurysm; embolization of the peripheral aneurysms.
(Note: Both types can accumulate thrombus inside)

4. Non-atherosclerotic lesions

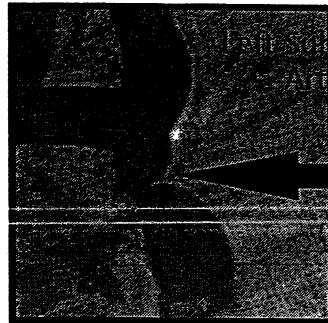
- a. Arteritis

- Can affect tibial and peroneal arteries, as well as the smaller more distal arterioles and nutrient vessels.
- Inflammation of arterial wall leads to thrombosis of vessel;
- Type: Buerger's disease (thromboangiitis obliterans)
- Associated with heavy cigarette smoking.
- Occurs primarily in men <40 yr old.
- Patients present with occlusions of the distal arteries.
- Rest pain and ischemic ulceration present

Study - Pulse of a
medial arch - arm

b. Coarctation of the aorta

- One of several congenital anomalies of arterial system.
- Congenital narrowing or stricture of thoracic aorta ; but may affect abdominal aorta.
- Example: (see arrow)



Left subclavian
Artery

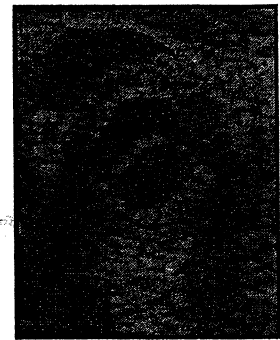
Coarctation of aorta

CLINICAL FINDINGS:

- Hypertension due to decreased kidney perfusion.
- Manifestations of LE ischemia (e.g., decreased pulses and/or segmental pressures).

c. Dissection

- Can affect aorta and peripheral arteries
- Distinguishing ultrasound feature: thin membrane dividing the arterial lumen into 2 compartments
- Media is weakened; intima develops tear through which blood leaks into the media (false lumen)
- Flow velocities differ in each lumen
- Aortic dissections which can extend to iliacs, may occur consequent to hypertension or severe chest trauma
- Complications: stenosis, occlusion or thrombosis



d. Vasospastic disorders: Refer to Chapter 3

e. Entrapment syndrome: Refer to Chapter 14

1. CAPABILITIES

- a. Help confirm diagnosis /approximate location of arterial occlusive disease.
- b. Indicate severity of the occlusive process.
- c. Combined with Doppler segmental pressures

2. LIMITATIONS

- a. Patients with casts or extensive bandages.
- b. Waveforms may be affected by ambient temperature.
- c. Uncompensated congestive heart failure may result in dampened waveforms.
- d. Unable to discriminate stenosis from occlusion.
- e. Technically dependent test.

3. PATIENT POSITIONING

- a. Supine with the extremities at the same level as the heart. To decrease influence of hydrostatic pressure.
- b. The patient's hip is externally rotated, knee slightly bent.
- c. Other positions: Rt/Lt lateral decubitus position, or prone.

4. PHYSICAL PRINCIPLES

- a. The Doppler effect:
 - When a wave is reflected from a moving target, the frequency of the wave received is different (Doppler shift) from the transmitted wave.
 - This effect occurs with relative motion between the source and the receiver of the sound.
 - Blood is moving target; transducer is stationary source.

- b. Continuous-wave (CW) Doppler is utilized.

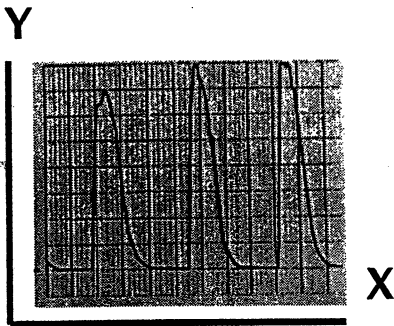
The reflected frequency is higher/lower than the transmitted frequency, depending on direction of flow.

- c. Types of Doppler Velocimetry:

Analog: Employs a zero crossing frequency meter, to display the signals graphically on a strip chart recorder.

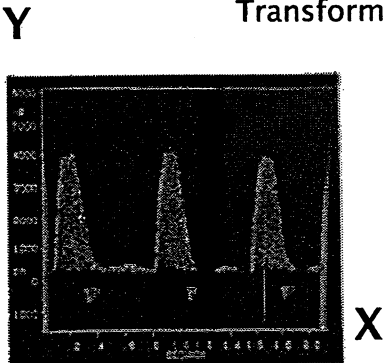
ZERO CROSSING FREQUENCY METER

- Circuitry counts each time the input signal crosses through zero (the baseline) within a time span.
- High frequency waves have many oscillations; low frequency waves have few.
- Direction of blood flow varies during the cardiac cycle.
- Machine estimates frequencies present in reflected signal and displays them.



- * Has acceptable accuracy
- * Drawbacks include: noise, less sensitivity, high velocities are underestimated; low velocities are overestimated.
- * Most equipment does a self calibration when the system is activated.

Spectral analysis: Individual frequencies displayed by Fast Fourier Transform (FFT) method.



- * More commonly used during duplex evaluation
- * Time displayed on the horizontal axis (X); frequency shifts displayed on the vertical axis (Y).
- * Free of many of the analog recording drawbacks.

5. TECHNIQUE

- a. Acoustic gel applied; 8 -10 MHz Doppler probe utilized.
- b. Upper extremities: Doppler velocity wave forms recorded from the following arteries bilaterally:

- Subclavian
- Axillary (axilla)
- Brachial (at elbow-antecubital fossa)
- Radial (thumb side, at wrist)
- Ulnar (5th finger side, at wrist)

- c. Lower extremities: Doppler velocity wave forms recorded from the following arteries bilaterally.

- Common femoral (CFA)
- Superficial femoral (SFA)
- Popliteal
- Posterior tibial (PTA) (medial malleolus)
- Dorsalis pedis (DPA) (top of foot)
- Peroneal (if necessary) location: lateral malleolus

- d. The audible AND wave form qualities are observed, documented, and combined with Doppler segmental pressures.

- e.

POTENTIAL SOURCES OF TECHNICAL ERROR

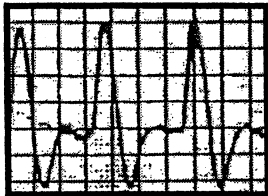
- Improper probe position
- Inadvertent probe motion
- Incorrect angle of incidence
- Inadequate amount of gel
- Excessive pressure on the probe tip
- Insufficient period of rest before testing

6. INTERPRETATION—Qualitative

a. Normal signals:

TRIPHASIC

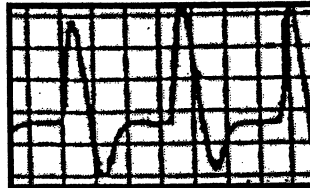
- Rapid upslope
- Sharp peak
- Rapid downstroke
- Flow reversal
- Resumption of forward flow
- Example: UE, LE arteries



b. Abnormal signals:

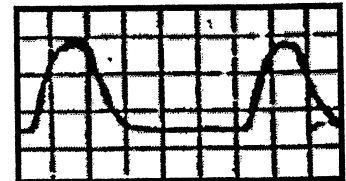
BIPHASIC

- Rapid upslope
- Sharp peak
- Fairly rapid downstroke
- Flow reversal
- No resumption of forward flow
- Also considered normal in some patients



MONOPHASIC

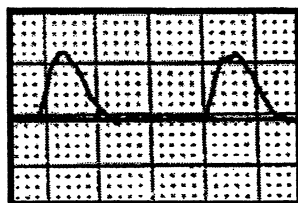
- Slow upslope
- Rounded peak
- Slow downstroke
- No reversal



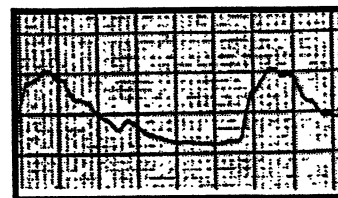
- Essential to observe for deterioration in the signals from one level to the next.
- A monophasic/dampened (pulsatile) signal is often obtained proximal to an obstruction. (In this example there is no diastolic flow.)
- Well collateralized occlusions can appear similar to flow distal to a stenosis.

Vasodilatation of the distal vessels often occur with proximal obstruction, reducing the pulsatility, causing the signals to have lower resistant (steady) flow quality

Miscellaneous Examples:

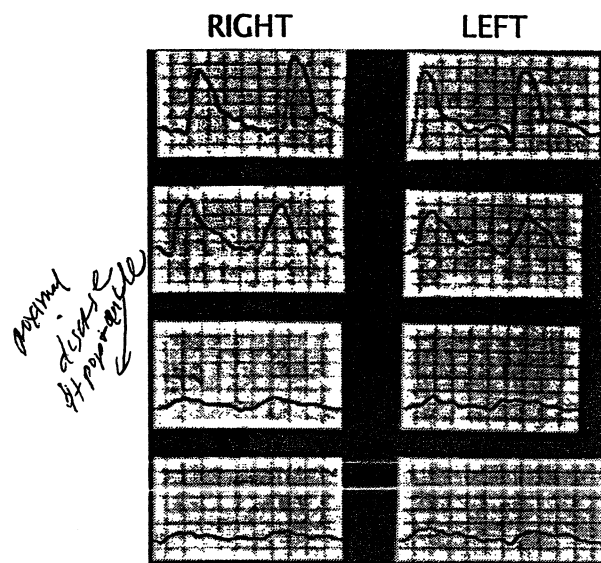


- Proximal to Obstruction (monophasic, pulsatile)

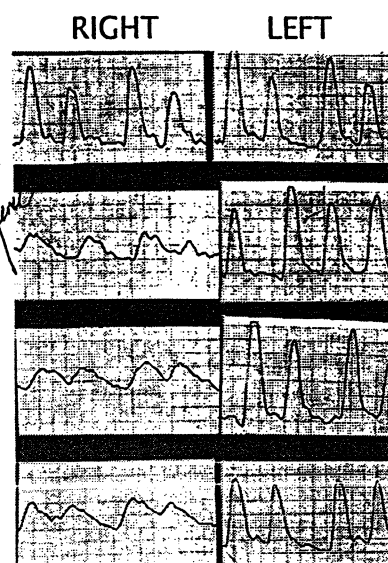


- Distal to Obstruction (monophasic, more steady)

Not triphasic - aortic pop



Waveforms only: suggestive of bilateral AI, multi-level occlusive disease

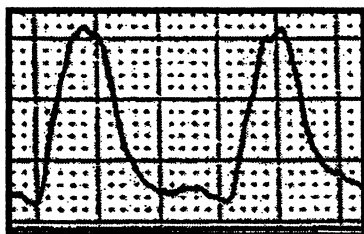


Waveforms only: suggestive of R. fem-pop occlusive disease and some element of aorto-iliac disease

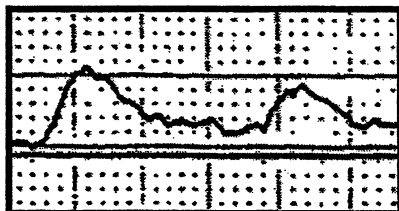
f. Upper extremity signals:

- Subclavian A: high resistant, multiphasic flow. Proximal occlusion or critical stenosis will make the signal more monophasic.
- Strandness and Sumner describe arteriovenous shunts in the skin of fingertips that cause flow patterns in hand to be tremendously variable; e.g., continuous signals can be found in the brachial, radial, and ulnar arteries in the relaxed and warm patient.

g. Doppler waveforms post-exercise: (If obtained)



- **NORMAL:** pre-exercise wave form qualities are maintained and/or augmented. No reverse component. Usually just pressures obtained post-exercise.



- **ABNORMAL:** Slow upstroke with more rounded peak, slow downstroke, no reverse component.

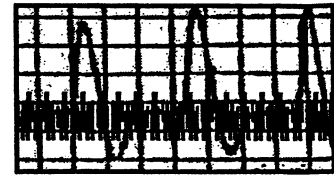
h. Absent Doppler signals may suggest occlusion or a pre-occlusive vessel (string sign.) Analog Doppler not capable of portraying velocities of less than 6 cm/sec.

i. Troubleshooting:

- *Recorder stylus not recording?* Check that proper test selection and/or probe type has been made.

*at 50%
on waveform
distal
response*

- *"60 cycle" noise on tracing?* Decrease gain; turn system off/on; increase filter; try another plug receptacle.

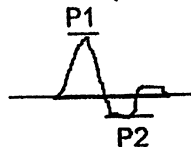


- *Recording stylus "stuck" at lower or upper portion of tracing?* Activate "re-set" control to re-center.

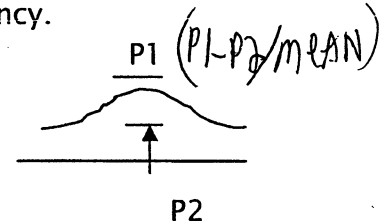
7. **INTERPRETATION—Quantitative** (using spectral analysis)

- a. **Pulsatility Index (PI):** Calculated by dividing the peak-to-peak frequency difference (P1 - P2) by the mean (average) frequency.

NORMAL



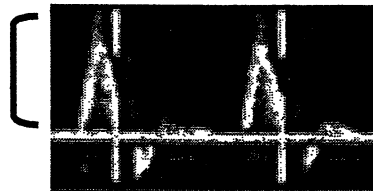
ABNORMAL



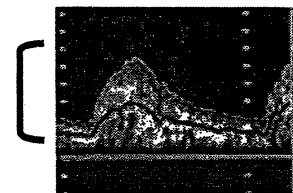
- Differentiate inflow from outflow disease. (Example: aortoiliac from femoral disease). *What is going on proximally*

- b. **Acceleration Time:** helps to differentiate inflow (aorto-iliac) disease from outflow (S-femoral) disease.

- Based on principle that proximal arterial obstruction results in a slowing of the time interval between the onset of systole to the point of maximum peak. *accelerated proximal distal - slow on stroke*
- Example criteria: An acceleration time of >133 msec suggests presence of proximal iliac disease.



Normal (CFA)



Abnormal (CFA)

1. CAPABILITIES

- a. Assess presence/severity of arterial disease.
- b. Combined with Doppler velocity or volume pulse waveforms.

2. LIMITATIONS

- a. Cannot discriminate between stenosis and occlusion, precisely localize area of obstruction, nor discriminate between CFA and external iliac disease
- b. Calcified vessels (medial calcinosis) render falsely elevated Doppler pressures; e.g., diabetics, end stage renal disease.
- c. Uncompensated CHF may result in decreased ankle/brachial indices.
- d. Artificially elevated high thigh pressures when narrow cuff used on thigh.
- e. Difficult to interpret in presence of multi-level disease.

3. PATIENT POSITIONING

- a. Patient should rest 20 minutes prior to the exam, especially when vascular disease present.
- b. Supine, with legs about same level as heart, so that hydrostatic pressure (which affects a sitting patient) cannot affect the BP measurements.

4. PHYSICAL PRINCIPLES

Refer to "Doppler Velocity Waveforms" evaluation.

REMINDER about cuff artifact:

- If the cuff is too large for a limb segment, BP is falsely lower
- If the cuff is too narrow for a limb segment, BP is falsely higher

5. TECHNIQUE

a. Appropriately sized blood pressure cuffs applied to both extremities.

- Place cuff "straight" on the extremity site, not encircling any bony prominence.
- Should fit snugly.
- Ideally the cuff bladder should be placed over the artery.
- Appropriate technique ensures the bladder inflation transmits pressure quickly into the tissue to compress the artery.
- Width of cuff should be about 20% greater than diameter of limb

b.

Four Cuff Method: (bilateral)

- Brachial (upper arm)
- High Thigh
- Above the knee (AK)
(low thigh)
- Below the knee (BK)
(calf)
- Ankle
- Size: 12 x 40 cm with longer
cuff bladders for thighs

c.

Three Cuff Method: (bilateral)

- Brachial (upper arm)
- One thigh cuff (19 x 40 cm)
- Below the knee (calf)
- Ankle
- Size: 12 x 40 cm; with ex-
ception of thigh cuff

d. Difference between the 4 cuff and 3 cuff method is:

- Two thigh cuffs (provide proximal and distal pressure measurements) but artifactually elevated BP's are obtained.
- The 3 cuff method utilizes one large thigh cuff (high on the thigh), providing a more accurate pressure reading.

e. Optimization of the Doppler signal:

- Utilize an 8-10 MHz Doppler frequency probe.
- Angle the CW Doppler probe 45-60 degrees to the skin.
- Due to vessel angulation, probe angle behind the knee may be closer to 90 degrees to the skin.
- Angle the probe so blood flow moves antegrade (towards the probe).

- hand held
blood pressure
device*
- f. Using a hand held sphygmomanometer, or an automatic cuff inflator, segmental Doppler pressures are obtained in the following order bilaterally:

ORDER OF SEGMENTAL PRESSURES

- Brachial (upper arm) using brachial artery
- Ankle (Use PTA and DPA; peroneal A. only if necessary).
- Calf (BK) (Use the site - PTA or DPA - that had the highest pressure, for obtaining the pressure for this level)
- Above the knee (AK) (Same as calf; may need to use popliteal A. if difficult to obtain)
- High thigh (HT) (Same as above knee)
- If one large thigh cuff used, protocol same as for the calf. ✓

*know
order
&
why*

NOTE: You must start at the ankle and move proximally to eliminate the possibility of underestimating the systolic pressure measurement.

For example, the high thigh cuff is inflated to above systole, deflated slowly to determine the BP; cuff is fully deflated; then the next distal cuff (AK) is inflated. Because this is done quickly, there isn't enough time for the arterial blood to completely flow normally into the leg, so the obtained BP would likely be falsely lower than it should be.

Note: Usually, the higher of the two pedal Doppler pressures is used to obtain the remainder of the segmental pressures in the leg beginning with the calf level.

- g. Toe pressures are discussed later in this section.
- h. Hints regarding segmental blood pressures:

REMEMBER!

- Complete cessation of blood flow is required; cuffs inflated 20-30 mmHg beyond last audible Doppler arterial signal; OR
- Inflate the cuff 20-30 mmHg higher than the highest brachial pressure
- If pressure measurements must be repeated, the cuff should be fully deflated for about a minute prior to repeat inflation
- The systolic pressure is recorded as that pressure at which the first audible Doppler arterial signal returns

6. INTERPRETATION

- a. Ankle/brachial index (ABI) is calculated by dividing the ankle pressure by the higher of the two brachial pressures.
- b. Another term for this is ankle/arm pressure index (API).
- c. The following criteria may vary slightly from Vascular Lab to Vascular Lab.

<u>ANKLE/BRACHIAL INDEX</u>		
> 1.0	=	Normal
>0.9 - 1.0	=	Asymptomatic disease or mild arterial disease.
0.5 - 0.9	=	Claudication (moderate disease)
< 0.5	=	Rest pain (severe arterial disease)

50% of brachial pressure

Note: Incompressible vessels have falsely elevated and inaccurate pressures. An ABI of >1.3—1.5 is considered incompressible.

d. Additional observations:

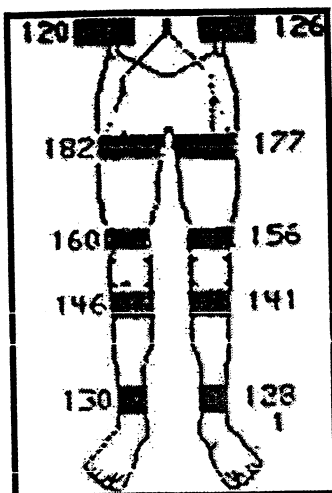
- Some authors feel an absolute ankle pressure (< 50 mmHg), rather than the ABI of .5 has stronger weight for predicting symptoms at rest.
- Strandness' work suggests that an ABI of $\geq .5$ represents single segment disease; <.5 suggests multiple lesions.

e.

- Segmental pressure drops of >30 mmHg (some authors suggest 20-30 mmHg) between two consecutive levels suggests significant obstruction.
- A horizontal difference of $\geq 20-30$ mmHg suggests obstructive disease at or above the level in the leg with the lower pressure.

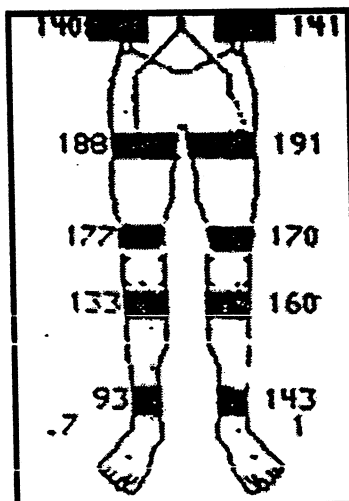
NOTE: The following examples are for self-assessment and are different from examples shown during slide presentation.

EXAMPLE #1



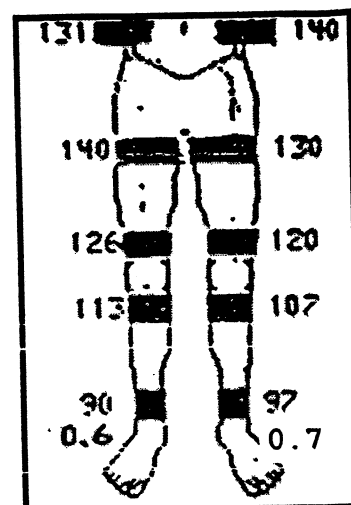
NORMAL

EXAMPLE #2



ABNORMAL

EXAMPLE #3



ABNORMAL

(Source of drawings is unknown)

f. Answers to the previous examples:

- #1 Normal
- #2 Femoral-popliteal and tibial vessel occlusive disease: >30mmHg gradient between the AK to calf and calf to ankle levels on the right; and 20-30 mmHg difference from right to left at calf and ankle levels.
- #3 Aorto-iliac (A-I) occlusive disease: low pressures at both high thigh levels only.

g.

(4) CUFF TECHNIQUE

- The high thigh pressure is normally ≥ 30 mmHg than highest brachial pressure.
- The AK and BK pressures should be at least the same as the highest brachial.

(3) CUFF TECHNIQUE

- Thigh pressure is similar to the highest brachial pressure.
- Does not allow for differentiation of HT to AK pressures.

Normal for high pressure to resist to brachial

h. Pressure levels for foot ulcer healing:

- Toe pressures of ≤ 30 mm Hg are evident in foot and toe ulcers (lesions) that failed to heal.
- Ankle pressures cannot always be relied upon; (e.g., diabetic patients) toe pressures may be more reliable.

EXERCISE

- Compare resting values to those obtained after exercise.
- Helps differentiate between true and pseudo-claudication.
- Determine presence/absence of collaterals.
- Contraindications include: shortness of breath, hypertension, cardiac problems, stroke, walking problems.
- TECHNIQUE** (After resting exam)
 - Patient walks on a constant load treadmill at a $\leq 12\%$ elevation.
 - Speed: 1.5 MPH for a maximum of 5 minutes or until symptoms increase to such severity that patient must stop.
 - Document: duration of walking, MPH, onset, location and progression of symptoms.
 - Post-exercise Doppler pressures obtained: both ankles (abnormal first), then higher brachial.
 - Post-exercise ABI's are obtained :
 - ◇ Immediately (Normal: ABI increases)
 - ◇ Abnormally, ABI decreases minimally or to a severe amount.
 - ◇ With drop after exercise, pressures obtained every two minutes until pre-exercise pressures are attained.

f. **INTERPRETATION**

INCORPORATES:

- Duration of exercise
- Length of time to recover.
- Pressure changes from pre to post exercise.

Single level Disease:

- Takes 2 to 6 minutes for the ABI's to increase back to resting levels after they dropped to low or unrecordable levels after exercise.

Multi-level Disease:

- Takes from 6 - 12 minutes for the ABI's to increase back to resting levels after they remained low or at unrecordable levels after exercise.

g. REACTIVE HYPEREMIA

- An alternate method for stressing the peripheral circulation. Used when patients: cannot walk long enough, use a cane or walker, have pulmonary problems; poor cardiac status, or other situations.
- Bilateral thigh cuffs (19 x 40) inflated to suprasystolic pressure levels (usually 20-30 mm Hg above the higher brachial BP) maintaining the pressure 3 to 5 minutes.

Produces ischemia and vasodilatation distal to the occluding cuffs.

- Upon release of cuff occlusion, ABI's are obtained.
- Normal limbs may show a transient drop of 17-34%.

Single level Disease

- $\leq 50\%$ drop in ankle pressure with reactive hyperemia.

Multi-level Disease

- $>50\%$ ankle pressure drop is seen

Treadmill testing is the preferable test because it produces a physiologic stress that reproduces a patient's ischemic symptoms.

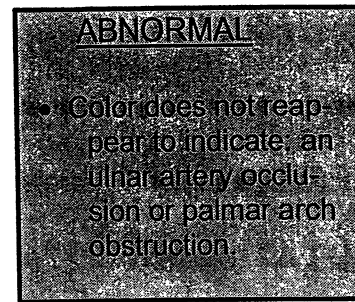
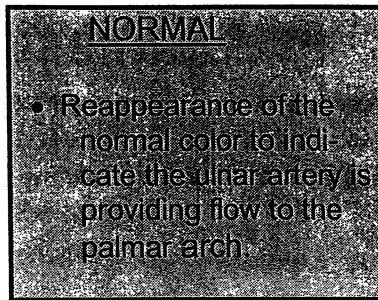
1. **CAPABILITIES** See previous "Lower Extremities"
2. **LIMITATIONS** See previous "Lower Extremities"
3. **PATIENT POSITIONING**
 - Supine with arms relaxed at the patient's sides.
 - Cuffs should be on snug, but not tight.
4. **PHYSICAL PRINCIPLES**

(Refer to previous "Doppler Velocity Waveforms" Evaluation)
5. **TECHNIQUE**
 - a. 12 x 40 cm cuff placed snugly on the upper arm; 10 x 40 cm cuff on the forearm bilaterally.
 - b. Brachial artery used to obtain upper arm BP; radial and ulnar arteries used to obtain the forearm pressure.
 - c. Pressures are combined with Doppler velocity wave forms from the sites previously described.
 - d. **Digits** are discussed in Chapter 9
6. **ALLEN TEST** to evaluate patency of the palmar arch.

Technique:

 - a. With manual compression of radial A by the technologist, patient clenches ipsilateral fist (<1 min.) inducing pallor.
 - b. With manual compression of the radial artery continuing, patient is asked to relax the hand.

c. Interpretation:



d. Limitations:

- Excessive dorsiflexion of wrist may compress radial and ulnar arteries leading to a false positive test.
- If hand is opened, fingers forcibly extended; the skin over palm can be stretched, (could lead to pallor due to compression of small vessels).

e. Documentation:

- Use a PPG on the index finger to document arterial pulsations before and after the "clenched fist."
- Although more difficult to compress, the ulnar artery can be evaluated similarly to assess radial artery flow.

7. INTERPRETATION

- a. A 15-20 mm Hg difference from one brachial pressure to the other, suggests a >50% stenosis of subclavian artery and/or the vessel under the cuff. (Refer to Chapter 21)
- b. A >15-20 mmHg drop from upper arm to forearm suggests:
 - A brachial artery obstruction distal to the upper cuff
 - Obstruction in both radial and ulnar arteries.
 - Obstruction in single forearm artery which has decreased pressure.
- c. A difference of ≥ 20 mm Hg between radial and ulnar pressures suggests obstruction in vessel with lower P.
- d. See Chapter 9 (Digits) for additional information.

1. CAPABILITIES

Helps determine whether impotence is related to peripheral vascular insufficiency.

2. LIMITATIONS

- a. High level of anxiety or antagonism.
- b. Duplex needed for velocities and anatomic conditions.
- c. Sensitivity to injectable medication; and/or anticoagulation therapy may prevent use of the injection component.

3. PATIENT POSITIONING

- a. Supine, with appropriate draping done to maintain privacy.

4. PHYSICAL PRINCIPLES

- a. Doppler previously described.
- b. Plethysmographic techniques described in next section.
- c. Duplex scanning principles.

5. TECHNIQUE / INTERPRETATION: Non-imaging

Technique:

- a. Doppler velocity signals obtained bilat. (CFA, PTA, DPA).
- b. Appropriate BP cuffs placed on upper arms, and ankles (refer to LE arterial):
 - Calculate ABI (previously described, Chapter 5)
 - Poor arterial inflow to LE due to proximal obstruction may affect penile arterial inflow.
- c. Penile pressures obtained with Doppler or PPG as the "end point detector."
Probe placement: lateral or ventral. Cuff size: 2.5 cm X 12.5 cm
- d. Penile plethysmographic waveforms obtained as needed.

Interpretation:

- a. Evaluation of the Doppler waveform quality: CFA, PTA and DPA as described previously.

b.

PENILE / BRACHIAL INDEX

- Normal = ≥ 0.75
 - Marginal = $0.65 - 0.74$
 - Abnormal* = < 0.65
- * Consistent with vasculogenic impotence

- c. Reduced pressure highly suggestive of more proximal arterial disease (e.g., aorto-iliac: internal iliac arteries).
- d. Abnormal plethysmographic tracings show marked reduction in amplitude.

6. **TECHNIQUE / INTERPRETATION: Imaging**

Technique: (Use a 7 or 10 MHz transducer)

- a. Obtain informed consent
- b. Prior to injection of medication, the cavernous (cavernosal) arteries: are measured (A/P diameter) in transverse approach; PSV, EDV obtained.
- c. Specific medication is injected into lateral aspect, proximal shaft of penis to induce erection.
- d. After 1-2 minutes post-injection, the same measurements are obtained as before injection (see b. above)
- e. Dorsal veins flow velocity measured.
- f. Important: If rigid erection is maintained for up to 3 hours, patient must be instructed to contact his Urologist immediately to reverse the priapism.

Interpretation: (Varies in the literature)

- Normal: Diameter of cavernous arteries should increase post-injection.
- PSV measurements should increase.
Normal range: PSV approx. 30 cm/sec or higher. Anything less often considered abnormal.
- Dorsal vein velocities should not increase; an increase could suggest venous leak. Normal: < 3 cm/sec; Abnormal: > 20 cm/sec.

1. CAPABILITIES

- a. In combination with Doppler segmental pressures helps differentiate true claudication from non-vascular sources.
- b. Detect presence/absence of arterial disease while defining its functional aspects.
- c. Help localize the level of obstruction.
 - Assessment of follow-up treatment.
 - PPG mainly used for evaluation of digits and penile vessels.

2. LIMITATIONS

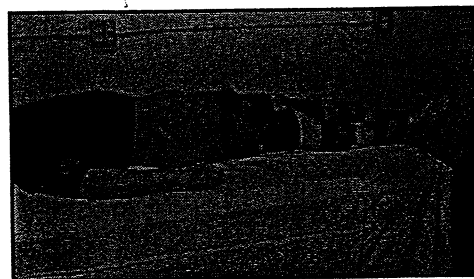
- a. Cannot be specific to one vessel. Tracing reflects all arterial flow beneath the cuff.
- b. Cannot discriminate between major arteries and collateral branches. (Venous influence not completely eliminated).
- c. Difficult to perform volume plethysmography on obese patients.

3. PATIENT POSITIONING

Most exams can be done with the limbs in a resting position (patient supine). The patient can be sitting for evaluation of upper limb digits.

4. PHYSICAL PRINCIPLES

Volume (air) Plethysmography
(Measurement of volume change)



- a. Pneumatic cuffs are placed around specific levels of the extremities or digits.

- b. A measured amount of air is sequentially inflated into a cuff to pressure ranging from 10 to 65 mm Hg, depending on cuff size.
- c. As arterial flow moves under the cuff, momentary volume changes in the limb segment occur.
- d. These changes beneath cuff are converted to pulsatile pressure changes within the air-filled cuff bladder.
- e. A pressure transducer converts the pressure changes into analog waveforms for display on strip-chart recorder.

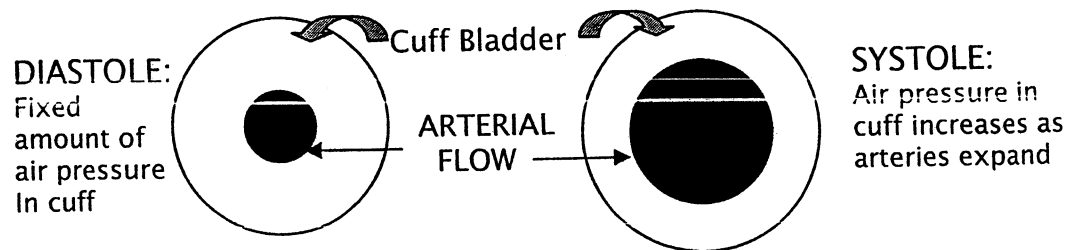
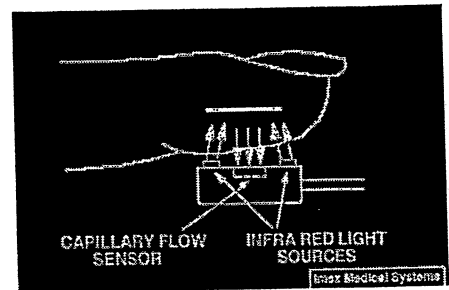


Photo-plethysmography (PPG)

- a. Consists of transducer, amplifier, strip-chart recorder.
- b. Detects cutaneous blood flow, rather than truly measuring volume change
- c. Photocell consists of light emitting diode and photo-sensor.
- d. Diode transmits infrared light into subcutaneous tissue with backscattered light reflected back to the adjacent photo-sensor.
- e. The cutaneous blood flow determines the reflection.
- f.  Blood attenuates light in proportion to its content in tissue. Increased blood flow results in decreased reflection. However, that's displayed as an increase/positive deflection on the waveform.



5. **TECHNIQUE**

Volume Plethysmography

- a. Patient supine, with heels slightly elevated on cushion.
- b. Using a 3 or 4 cuff method, appropriately sized pneumatic cuffs applied snugly to: thigh, calf and ankle bilaterally.
- c. Machine performs a self-calibration when activated.

- d. Bilateral brachial pressures obtained during the Doppler segmental pressure exam.
- e. Appropriate amount of air used to bring cuff pressure to predetermined levels. Begin with upper part of extremity, moving distally. Record at least 3 pulse cycles.
- f. Artifacts not uncommon due to improper cuff application.
- g. Similar "Gain" setting is maintained throughout the study. If a different setting is used, note on the recording paper.
- h. Volume plethysmography and pressures are complementary

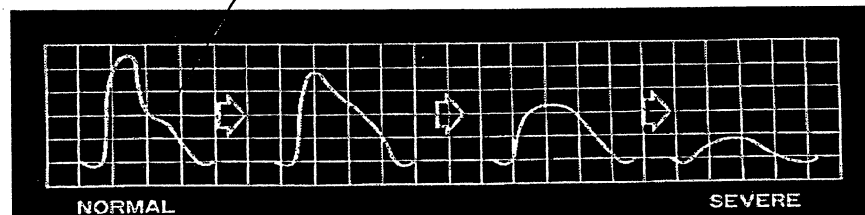
Photo-plethysmography (PPG)

See DIGITS, Chapter 9

6. INTERPRETATION

Both "volume" and "photo" plethysmography are evaluated using qualitative criteria as described by DE Strandness, MD and J Raines, PhD. Major descriptions follow:

- a. Normal: fairly rapid upslope, sharp systolic peak with reflected wave.
- b. Mildly abnormal: Sharp peak, absent reflected wave, downslope is bowed away from baseline.
- c. Moderately abnormal: Flattened systolic peak, upslope and downslope more delayed; reflected wave (notch) absent.
- d. Severely abnormal: Low amplitude, or may be absent.

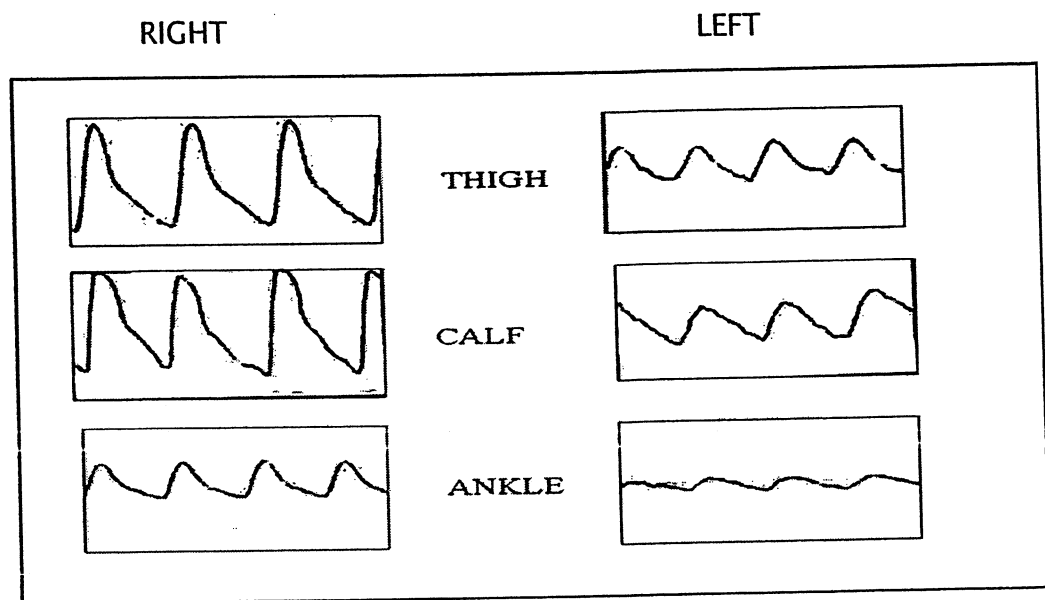


ADDITIONAL NOTES:

- e. Abnormal waveforms always reflect hemodynamically significant disease proximal to level of tracing. Severity generally underestimated.
- f. Reduced amplitude with no changes in the contour is likely to reflect insignificant disease, unless it is unilateral.



g. EXAMPLE STUDY:

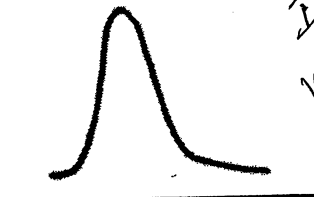


Abnormal plethysmographic waveform quality suggests poor arterial flow at all levels on the left. There is normal waveform quality seen on the right at all levels.

- h. A fair waveform quality may accompany abnormal segmental pressures because collaterals can underestimate significance of obstruction based upon plethysmography.



Normal



Based on collaterals

If no mtr vessel pressure is filling w/ collaterals

Note: Amplitude similar; waveform qualities different.

- i. Additional qualitative criteria found in the next chapter.

TROUBLESHOOTING

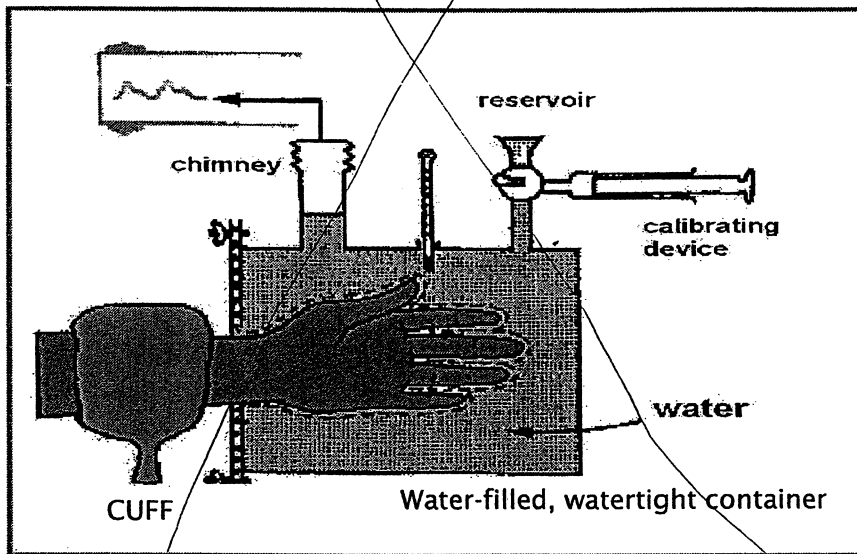
- j.

- *Can't center the stylus?* Check the "mode." AC mode for arterial; DC mode for venous.
- *Stylus wandering on paper?* Activate the "re-set" control; and be sure correct exam function is selected.
- *Unable to obtain clean waveform?* Reapply the PPG. Patient tremors make an exam nearly impossible.
- *No tracing?* Check the exam mode, paper, and connection points.

7. DISPLACEMENT PLETHYSMOGRAPHY:

- a. Any change in volume of the enclosed part will displace an equal amount of water. Displacement is measured by the height of the water in the chimney.
- b. Accurate for detecting changes in limb volume.
- c. Pulse plethysmography refers to transient changes in (limb) volume related to the pulse by pulse activity of the left ventricle; the (body) part expands when arterial inflow exceeds venous outflow.

In the example below, hand is inside a surgical rubber glove (as depicted by the dotted/dashed line). The water temperature is constant.



From: Noninvasive Diagnostic Techniques in Vascular Disease Eugene F. Bernstein, M.D., 3rd edition, C.V. Mosby 1985

1. CAPABILITIES

- a. Help detect presence of arterial disease.
- b. Differentiate fixed arterial obstruction from vasospasm.
- c. Assess effects of treatment.

2. LIMITATIONS

- a. Quality of tracings greatly affected by the vasoconstricted state of arteries before testing begins. (Has patient: come in from cold weather? been smoking? nervous?)
- b. With **volume plethysmography**, cuffs applied too tightly, can obliterate or diminish the pulse wave forms.
- c. With **photo plethysmography** improper contact with skin surface will cause poor results.
- d. Bandages that can't be removed; ulcerations and/or gangrene may prevent placement of cuff or photocell.
- e. Patient with extremity tremor.

3. PATIENT POSITIONING

- a. Toe evaluation: supine with some elevation of head.
- b. Finger evaluation: sitting with arms resting on pillow placed on patient's lap.

4. PHYSICAL PRINCIPLES

The principles for volume and photo plethysmography have been discussed previously.

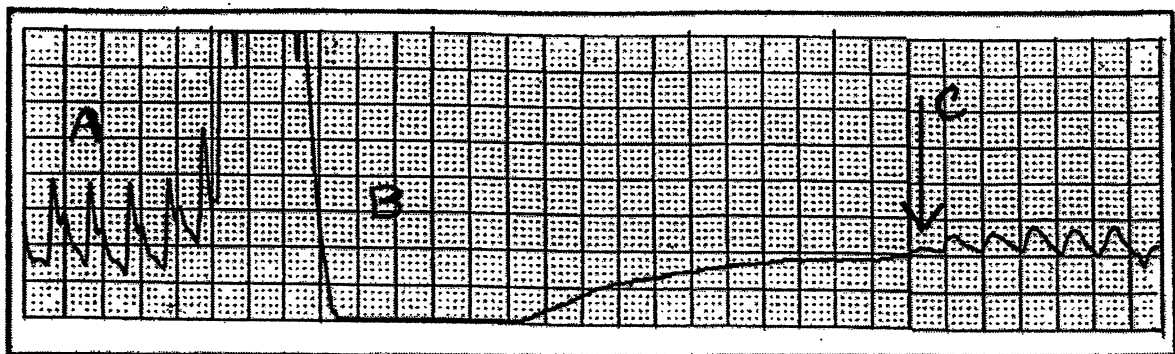
5. TECHNIQUE—TOES

- a. Patient should be kept as warm as possible.
- b. The study is done in combination with a complete LE arterial exam, or a limited version such as an ABI.

- c. An appropriately sized cuff, the width at least 1.2 times that of the toe, (2.5 - 3 cm) is applied to base of great toe.
- d. Equipment does a self calibration when activated.
- e. Plethysmographic waveforms are recorded for all toes as previously described. (If poor waveforms, warm the toes).
- f. Usual method for obtaining great toe pressures is PPG: (photocell as end point detector). METHOD FOLLOWS:

- Digit cuff placed at base of great toe.
- Photocell securely attached to plantar side of toe using double-stick tape or velcro strap.
- Pulses recorded; paper speed=5mm/sec. (A)
- While pulsations are recorded, cuff is inflated to 20-30mmHg above highest brachial P.
- No pulsations should be seen. (B)
- Cuff is slowly deflated, watching for return of first pulse to define the pressure level. (C)
- Can also be with volume plethysmography.

EXAMPLE:



The increase in toe volume (rise from A to B) secondary to obstruction of venous outflow. (Plethysmography measures all volume changes).

6. **TECHNIQUE: Fingers without cold stress**

- a. Upper extremity arterial study is completed initially.
- b. Evaluate Doppler signals (triphasic, biphasic, monophasic) and obtain pressures.
- c. Doppler exam of the palmar arch, to verify patency.
- d. Apply finger cuffs (sizes: 2 - 2.5 cm)
- e. Pressures and waveforms obtained (using similar method described for toes).

7. **TECHNIQUE: Fingers with cold stress**

Performed in cases of symptoms occurring due to **cold sensitivity**, (Toes can also be evaluated).

- a. After resting study is performed, hands are immersed in ice cold water for 3 minutes if possible.
- b. Following cold stress, waveforms and pressures are obtained: immediately after and five minutes after.
- c. Document patient symptoms, skin color observations and other pertinent findings on the report form.

8. **INTERPRETATION**

a. Normal waveform qualities:



- Sharp upstroke during peak systole
- Prolonged down-stroke with notch (reflected wave) approximately half way down.
- Amplitude is usually greater than toe tracings

b. Abnormal Obstructive waveform qualities:



- Occlusion located anywhere proximal to the tip of the finger causes the pulse to assume an "obstructive" form.
- Slow upslope to rounded peak.
- Downslope that bows away from baseline.

c. Abnormal Peaked waveform qualities:



- Upslope is slower than normal.
- Sharp, (anacrotic) notch is present
- Reflected wave located high on the downslope.
- Has characteristics of both normal and the obstructive waveforms.

d. Other observations:

- Organic (fixed) obstructive disease has abnormal Doppler arterial signals, systolic pressures and PPG tracings.
- Functional (intermittent) obstructive disease has normal Doppler arterial signals, systolic pressures, and/or PPG tracings; but abnormal findings after cold stimulation.
- Sumner and Strandness described the "peaked" pulse characteristically seen in the digit pulse contours of patients with Raynaud's phenomena.

Following cold immersion, abnormal cold sensitivity is likely if the amplitude fails to return to baseline levels within 5 minutes.



Resting



Immediately
after cold immersion



Five minutes after
cold immersion

e.

DIGIT PRESSURE MEASUREMENTS

- UE digits: Finger/brachial indices 0.8-0.9
- LE digits: Toe/brachial Index (TBI)
 - ◇ Normal toe P's vary from 60-80% of brachial P.
 - ◇ The presence of artifactually high ankle pressures from arterial calcinosis usually negates a toe/ankle P index.
 - ◇ Digital artery occlusion: severely reduced pressure.

1. CAPABILITIES

- Wound healing and amputation level determination.
- Reflects tissue oxygen tension and relies on a balance between oxygen supply and consumption.

2. LIMITATIONS

- Inability to keep electrode fairly flat on the skin surface.
- Electrode placement must be on intact skin.

3. PATIENT POSITIONING

Patient supine, head slightly elevated; room is warm

4. PHYSICAL PRINCIPLES

- a. Electrode heats skin to 45° C.
 - Blood flow increases
 - Lipid layer in fatty tissue melts
 - More "O₂" escapes through skin, measured by sensor in electrode
- b. As "O₂" escapes, a chemical reaction occurs in electrolyte solution, located between skin and electrode surface.
- c. Electrode converts chemical reaction to a "current" reading which is converted to a pO₂ reading in mmHg.



5. TECHNIQUE

- a. Skin cleansed with alcohol wipe; air dried.
- b. Airtight self-adhesive fixation ring placed on the skin
- c. Few drops of electrolyte solution put inside ring.
- d. Electrode/sensor is affixed to fixation ring.

- e. Reference reading (upper lat. chest) obtained first; followed by specific sites (e.g., near toes, BK, AK)
- f. After manual calibration, it takes approximately 15-20 minutes/site to obtain "pO₂" reading.

6. **INTERPRETATION**

- a. Normal values vary in literature and with manufacturer. Healing should occur with a pO₂ reading of 70-80 mmHg;
- b. Borderline healing: 30-40 mmHg.
- c. Non-healing (poor values) = 10-15 mmHg. This number also varies.

1. CAPABILITIES

- a. Localize stenosis / occlusion; evaluate degree of stenosis.
- b. Determine the presence/absence of aneurysm.
- c. Post-op study: hemodialysis access or arterial bypass graft.
- d. Detect AVF's or other unusual abnormality.

2. LIMITATIONS

- a. Limited access to extremity (e.g., dressings, skin staples or sutures, open wounds, IV site).
- b. Pertaining to hemodialysis access grafts:
 - Graft angulation
 - Difficult to adequately evaluate the outflow vein in an obese patient.

3. PATIENT POSITIONING

- a. Patient is supine with small pillow under head.
- b. Extremity close to the examiner.
- c. Arm is at a 45 degree angle from the body, and externally rotated. ("pledge position")

4. PHYSICAL PRINCIPLES

- a. Duplex scanning: combination of real-time B-mode imaging (gray scale evaluation) and Doppler spectral analysis.
- b. Doppler Color Flow Imaging: Doppler information is displayed on image after evaluated for phase (direction toward or away from transducer) and its frequency content (hue or shade of the color).

Brightness

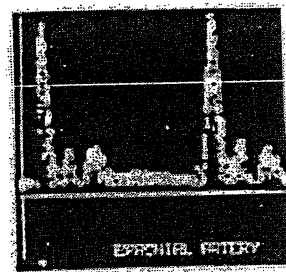
- c. Sample size for acquiring pulsed Doppler information is usually 1-1.5mm. Size is increased incrementally if needed.
- d. In-depth coverage of this process is covered in the morning "Physics Section."

5. **TECHNIQUE: Native Arteries**

- a. Utilize a 7 or 5MHz linear array transducer.
- b. Neck vessels identified, with attention given to innominate artery on the R. The LCCA branches off the Arch.
- c. Color/duplex scanning is also used to evaluate the following:

Subclavian
Axillary
Brachial
Radial
Ulnar
Palmar arch (if needed)

Example UE arterial signal →



- d. Uncommon for arteries in the upper extremities to become stenotic. Main use is evaluation of dialysis access grafts.

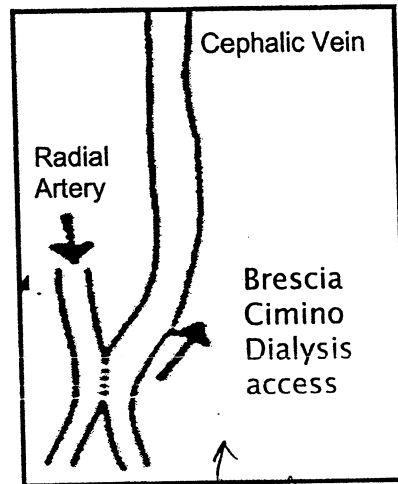
6. **TECHNIQUE: Hemodialysis Access**

arbitrary flow augmentation

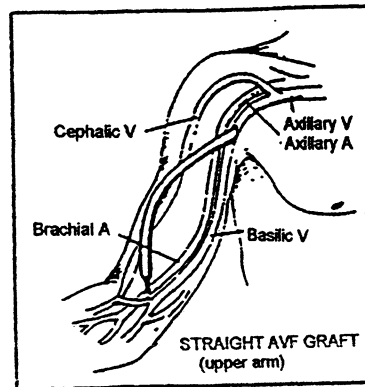
- a. Auscultate the access for bruit and/or palpate for a "thrill" (vibration). A patent dialysis access, as well as a stenotic one can produce a "thrill."
- b. Utilize a 7 or 5 MHz linear array transducer.
- c. Evaluate dialysis access grafts as follows:
 - Inflow artery
 - Arterial anastomosis
 - Continue through the body of the graft
 - Observe for aneurysm, puncture sites, peri-graft fluid.
 - If color available, observe the image for flow changes, turbulence, flow channel changes.
 - Venous anastomosis
 - Outflow vein

Note: Dialysis access (e.g. Brescia-cimino fistula) assessment sites include: inflow artery, anastomosis, outflow vein.

- d. Dialysis access examples include: Brescia-Cimino fistula; straight, looped synthetic grafts.



Low in the hand - Cephalic - Radial



Low in the arm

Graft - fistula - connection b/t artery & vein

7. Technique - General Observations:

- Longitudinal/transverse approaches used to evaluate gray scale for thrombus, stenosis etc.
- Doppler peak systolic velocities (PSV) obtained at appropriate sites and as needed. Volume flow may be evaluated.
- Documentation consists of video tape or prints.

8. **INTERPRETATION: Native Arteries**

NO QUESTIONS

a. Stenosis:

- Currently, no criteria for classifying disease as there is for LE.
- Normal peak systolic velocities vary widely with skin temperature changes. Doppler signal quality is usually triphasic.
- If a >50% diameter reduction is present, observe for characteristics of the "stenosis profile."

b. Occlusion:

- Observe for lack of Doppler signals (image and/or waveform) and the proverbial "thump" which is obtained proximal to occlusion.

c. Aneurysm:

- Dilation of the vessel from degeneration and/or weakening of the wall.

- Ulnar artery aneurysms can form in response to using the palm as a hammer.
- Subclavian aneurysms often are associated with embolization to the digits.

9. **INTERPRETATION: Hemodialysis Access**

- Identify/document if present (location, extent, and type) any aneurysmal changes, puncture sites, peri-graft fluid, thrombus.
- Peak systolic velocities (PSV) and end diastolic velocities (EDV) vary as to the type of access; normally both are elevated. Low PSV obtained in access graft could indicate arterial inflow problems.
- No standardized velocity criteria at this time. Follow-up studies provide specific comparison to previous studies on same patient.
- Venous anastomosis and outflow vein are most common sites for stenosis (likely caused by increased arterial pressure introduced to the vein and/or intimal hyperplasia).
- Other hemodynamic complications include:
 - Large blood volumes shunted from artery to lower resistant venous circulation, can increase venous return resulting in congestive heart failure.
 - A "steal syndrome:" the distal arterial blood flow is reversed into the lower resistant venous circulation and can cause pain on exertion, pallor and coolness of the skin distal to the shunt.

1. CAPABILITIES

- a. Determine presence/absence of >50% diameter reduction or occlusions.
- b. Determine presence/absence of aneurysms.
- c. Follow up of bypass grafts.
- d. To localize the stenotic lesion prior to balloon angioplasty.

2. LIMITATIONS

- a. Presence of dressings, skin staples, sutures; open wounds.
- b. Incisional tenderness, hematomas.
- c. Obesity - may be difficult to image vessels
- d. Calcific shadowing from diabetes and/or atherosclerosis.

3. PATIENT POSITIONING

- a. Patient supine with head on pillow.
- b. Extremity positioned close to examiner.
- c. Pt's hip minimally rotated externally, with knee flexed.
- d. Prone or lateral decubitus position may be required for assessment of popliteal artery.

4. PHYSICAL PRINCIPLES

Detailed presentation in the morning "Physics" session. The Doppler equation may be written as:

$$Df = \frac{2 F_0 V \cos \theta}{c}$$

- Df = Doppler shift frequency.
- Fo = Carrier frequency
- c = Speed of ultrasound through soft tissue (1540m/sec)
- V = Velocity of the moving reflectors
- Cos Ø = Angle

a. The number 2 represents two Doppler shifts:

- Red blood cell is first an observer of a stationary ultrasound field.
- Then acts as a wave source when the waves scatter from its surface

CALCULATING FOR VELOCITY:

$$V = \frac{c \cdot Df}{2 F_o \cdot \cos \theta}$$

- Df = Known value, measured by the duplex. Proportional to velocity of the source.
- V = Calculated value which represents velocity of flow (moving reflectors). To calculate, the frequency (Df) must be known.
- Cos Ø = Angle (Ø) determined from the image; cosine theta is calculated. (e.g., cos Ø of 90 = 0).

b. Source of error (solving for velocity) is the Doppler angle theta (Ø) which increases its nonlinear influence as the angle becomes closer to 90 degrees.

c. **Ideal Doppler Angle for Vascular exams is 60 degrees**, obtained centerstream, parallel to walls. Vessel tortuosity and other conditions, will cause the Doppler angle to be < 60 degrees, which is usable.
Not reliable: Doppler angle of > 60 degrees.

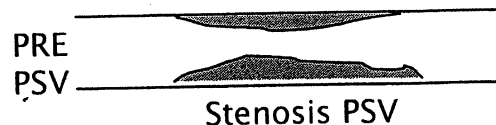
d. The effect on the Doppler shift frequency display when the carrier frequency is changed: Decreased transducer frequency = decreased Doppler waveform size (e.g., 8 MHz changed to a 4 MHz)

5. **TECHNIQUE: Native Arteries**

a. Scan with a 7 or 5 MHz linear array transducer for the following:

- Distal external iliac artery
- Common femoral artery (CFA)
- CFA bifurcation
 - Superficial femoral artery (SFA)
 - Profunda femoris

- S-femoral artery (proximal - distal)
 - Popliteal artery (proximal - distal)
 - Trifurcation
 - Anterior tibial artery (ATA)
 - Posterior tibial artery (PTA)
 - Peroneal artery
- b. Observe gray scale for plaque or other abnormalities.
- c. Observe color flow patterns.
- d. Obtain peak systolic velocities (PSV) from each major vessel (proximal, mid and distal if vessel is long) and more if disease suspected.
- Characterize the Doppler signals: triphasic (normal), biphasic, or monophasic (abnormal).
 - Observe wave form for any spectral broadening.
- e. If a >50% diameter reduction is suspected, obtain:
- Pre-stenotic PSV (approaching the stenosis).
 - PSV (highest) in stenosis.
- f. Post-stenotic turbulence signals.
- Document as previously described for UE.



6. TECHNIQUE: Bypass grafts

- a. Know type, location and age:
- Synthetic (e.g., Gortex)

Reversed Saphenous Vein Graft (RSVG)

- Small end is now proximal
- Large end is distal
- Vein valves stay open due to arterial flow pressure
- Branches are ligated

In-situ Vein Graft

- GSV stays in place
- Small end is distal
- Large end is proximal
- Prior to surgery, valves broken up with special instrument, branches ligated

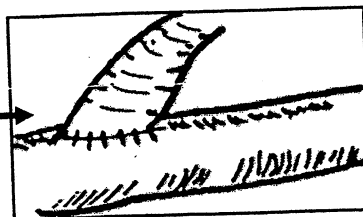
b. Protocol usually combined with ABI's.

c. Vein bypass graft evaluation includes: observation of gray scale, color flow and PSV's of the following:

- Inflow artery
- Proximal anastomosis
- Entire length of the vein bypass graft
- Distal anastomosis
- Outflow artery
- Also check for branches (that could form AV fistulas), valves, and/or other abnormalities.

d. Synthetic bypass graft evaluation include observations of gray scale, color image, PSV's of the following:

- Inflow artery
- Proximal anastomosis
- Mid graft
- Distal anastomosis
- Outflow artery



7. INTERPRETATION: Native Arteries

a. NORMAL:

- Doppler signals are triphasic; a change to biphasic can be significant; however some patients normally have biphasic flow without evidence of disease.

b. ABNORMAL FINDINGS:

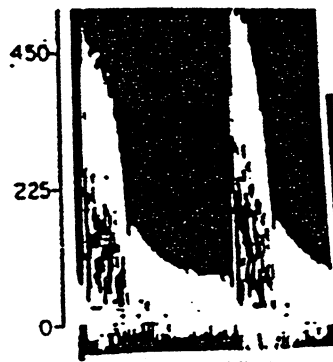
Compare stenotic PSV to pre stenotic PSV

- 2:1 ratio = $\geq 50\%$ diameter reduction
- 4:1 ratio = $\geq 75\%$ diameter reduction
- >400 cm/sec PSV = $\geq 75\%$ diameter reduction
- Never accept the value of "numbers" alone, post-stenotic turbulence should be present.

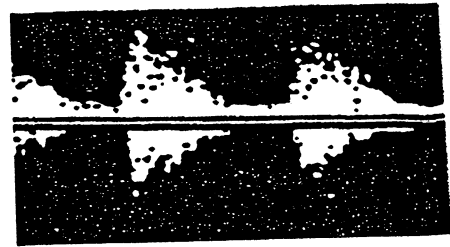
- Observe for occlusions; aneurysms.



Pre-stenotic Doppler spectra (monophasic and damped)



Doppler spectra obtained AT the stenosis (highest PSV's documented)



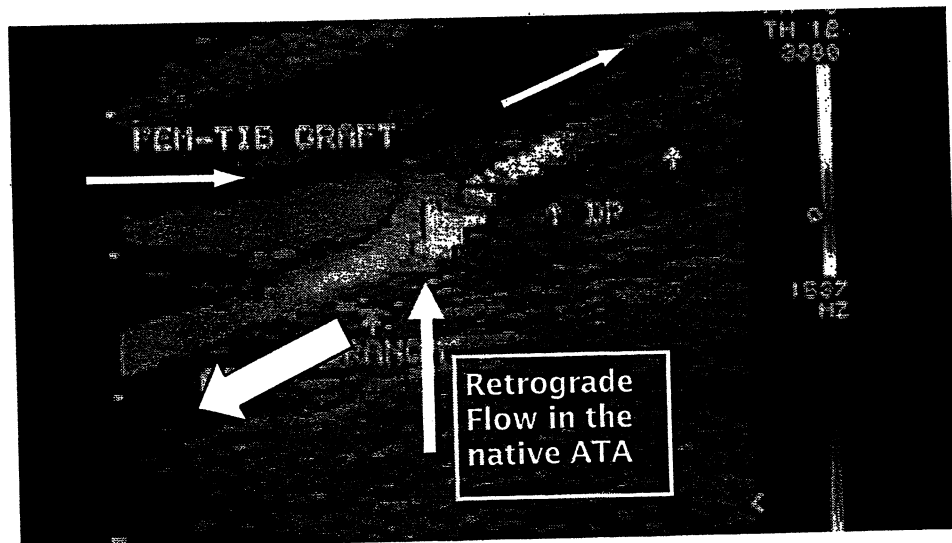
Post-stenotic turbulence (Evident distal to a > 50% stenosis)

8. INTERPRETATION: Bypass Grafts

a. NORMAL:

- As stated under "Native Arteries."
- Retrograde flow in the native artery may be evident at the distal anastomosis of a RSVG, which provides an additional source of collateral flow. (Retrograde flow results from a pressure gradient)

normal



b. ABNORMAL FINDINGS:

Know this ↓

Compare results with previous studies and observe for changes such as:

- Decrease of 30 cm/sec in any graft segment
- Reduced PSV's in smallest graft diameter that were greater previously
- Change from triphasic to biphasic signals
- Decrease in ABI of > 0.15
- Observe for post-complications such as: AV fistula (can siphon off graft flow); valve cusp
- Can also apply previous data used for native artery stenosis

Observe for post-complications such as: AV fistula (can siphon off graft flow); valve cusp

c. SYNTHETIC GRAFTS:

- 2-1/4-1*
- Can loosely apply the previous data to determine whether a $> 50\%$ diameter reduction exists.
 - Anastomotic sites should be evaluated well for aneurysm and/or stenosis.

d. Observe for graft occlusion.

9. **INTRAOPERATIVE MONITORING**

- Used for checking patency of the anastomotic sites.
- Evaluate any suspicious stenotic or turbulent areas that can occur in vein bypass grafts (e.g., valve cusp sites or suspected branch sites.)
- Further description of this protocol, see Carotid Duplex section.

1. CAPABILITIES

- a. Aorto-iliac vessels: evaluate vessels for stenosis, status of bypass grafts, aneurysmal disease.
- b. Renal artery: to document >60% diameter reduction.
- c. Kidney: to help in evaluation of nephrosclerotic disease, and transplants.
- d. Mesenteric arteries: to document significant stenosis, or evaluation of mesenteric bowel ischemia.
- e. Liver: suspected portal hypertension; pre/post liver transplants; other.

2. LIMITATIONS

- a. Size of patient
- b. Bowel gas
- c. Previous abdominal surgery (scar tissue)
- d. Shortness of breath and rapid respirations
- e. Patient in non-fasting state

3. PATIENT POSITIONING

- a. Supine, with minimal head elevation.
- b. Left lateral decubitus (LLD) for access to the right flank and the right lateral decubitus (RLD) for access to the left flank.
- c. Requires creative patient positioning and scanning approaches in order to optimize the study.

4. PHYSICAL PRINCIPLES

Previously described in "Duplex scanning / color flow imaging of the upper extremities."

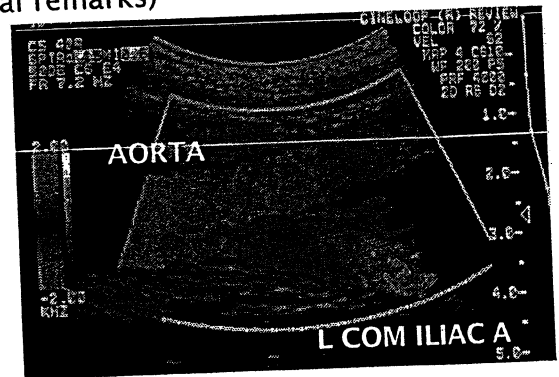
5. **TECHNIQUE: General Remarks**

- a. Transducer selection related to size of patient; most adult patients require 5, 3, or 2.25 MHz transducer.
- b. Combine longitudinal and transverse approaches; evaluate gray scale and color flow patterns, observing for aneurysm, plaque etc.
- c. Use longitudinal approach to assess Doppler signal PSV and EDV.

6. **AORTO-ILIAC ARTERIES: Technique/Interpretation**

TECHNIQUE: (See Technique - general remarks)

- a. Observe celiac and SMA
- b. Observe renal arteries
- c. Aorta: Proximal, mid, distal
- d. Common iliac A bilaterally
- e. Bil external iliac A Prox - distal
- f. Internal iliac artery bilaterally
- g. See technique section in Chapter 12



INTERPRETATION:

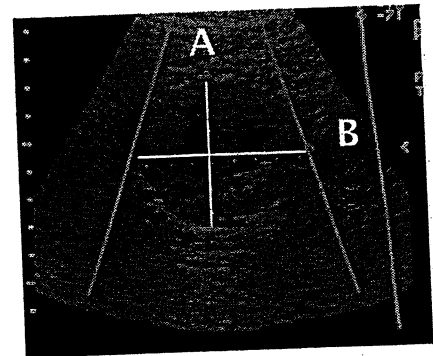
- a. **Aorto-iliac arteries - stenosis**

Please refer to the "Duplex scanning/ color flow imaging of the LE arteries."

- b. **Aorto-iliac arteries - aneurysm**

- A dilation > 3cm OR an increase in diameter of 50% greater than original artery.

A = 5.6 cm
B = 6.1 cm



- Majority of abdominal aortic aneurysms (AAA) are atherosclerotic and infrarenal (below the renal artery branches.)
- Type of aneurysm: true, (fusiform, saccular); false (thrombus present?)

Most frequent complication: Rupture of the aortic aneurysm;
embolization of the peripheral aneurysms.

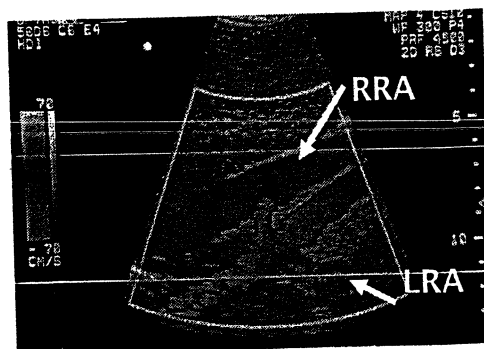
7. RENAL ARTERIES: Technique/Interpretation

Patient history: Many of these patients will present with hypertension (controlled or not). Many patients who have hypertension have **renovascular hypertension**, usually caused from renal artery stenosis (which can be secondary to atherosclerosis, fibromuscular dysplasia) or occlusion.

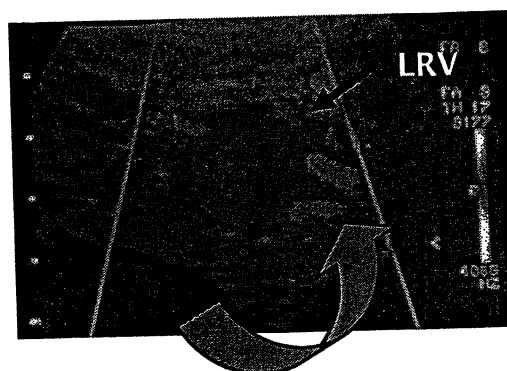
TECHNIQUE: (See Technique - general remarks):

- Celiac artery and SMA velocity data.
- Obtain aorta PSV near SMA level
- In transverse, locate renal arteries.

Left renal vein (LRV) is a landmark for identifying left renal artery.



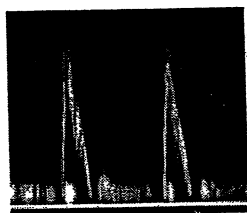
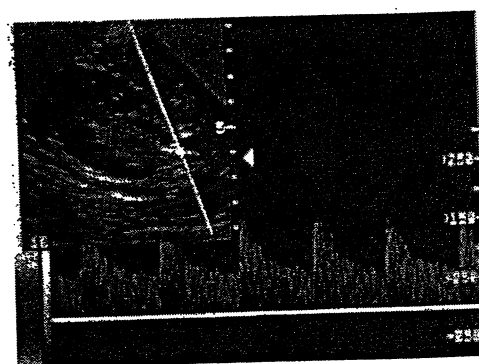
- Obtain kidney size (length) observe morphology bilaterally.
- Obtain PSV and EDV bilaterally of:
 - Renal artery - proximal, mid and distal bilaterally.
 - Upper / lower pole of the kidney in segmental arteries.



- Observe for secondary or accessory renal arteries.

INTERPRETATION:

- Renal arteries and kidney arteries are normally low resistant in quality as are: celiac, hepatic, and splenic arteries.
- The aorta is usually higher resistant in quality, as are a fasting SMA and IMA.



c.

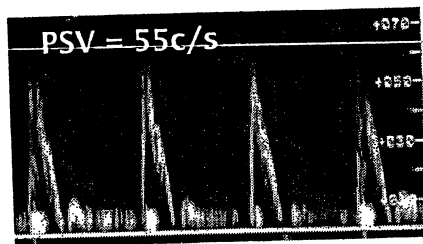
Renal to Aortic Ratio (RAR):

Highest Renal Artery PSV
Aorta PSV = 55 cm/s

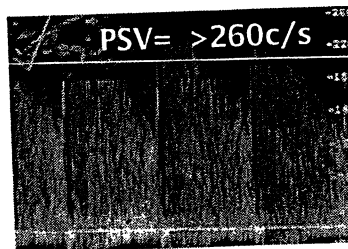
NORMAL = < 3.5
 ABNORMAL = ≥ 3.5 Suggests $\geq 60\%$ diameter reduction

Cannot use the RAR:
 1) If AAA detected 2) If aortic PSV $> 90\text{cm/s}$ or $< 40\text{cm/s}$

With these conditions, look for a renal artery PSV of 180-200cm/sec, that is also followed by post-stenotic turbulence.



AORTA



ABNORMAL RENAL A.

RAR:

$$\frac{> 260}{55} = > 4.7$$

INTERPRETATION - Kidney (Can also apply to renal arteries)

- Kidney arteries are normally low resistant in quality.
- Observe the kidney for morphologic abnormalities (cyst, cortex thinning, other defects.)
- Normal pole to pole length varies from 10 - 12cm (depends on patient size)
- Ratios such as the following are utilized:

END DIASTOLIC RATIO (EDR)

Parenchymal Resistance Ratio (PR)

$\frac{\text{End Diastolic V}}{\text{PSV}}$

Normal = > 0.2
 Abnormal = < 0.2

RESISTIVITY INDEX (RI)

$\frac{\text{PSV} - \text{EDV}}{\text{PSV}}$

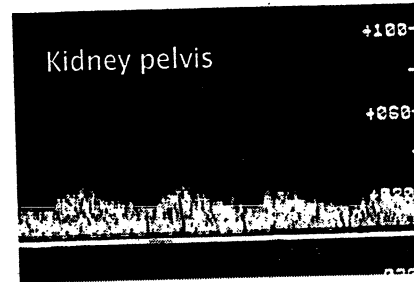
Normal = < 0.75
 Abnormal = ≥ 0.75

Abnormal calculations indicate an increase in distal resistance (e.g. nephrosclerotic disease).

Example data: PSV of 45cm/sec; EDV of 5 cm/sec.

- The EDR = 0.1
- The RI = 0.88

- e. Proximal high grade stenosis/occlusion of renal artery may result in dampened, weak Doppler signals distally, but still of low resistant quality. Another term for this is tardus-parvus. →



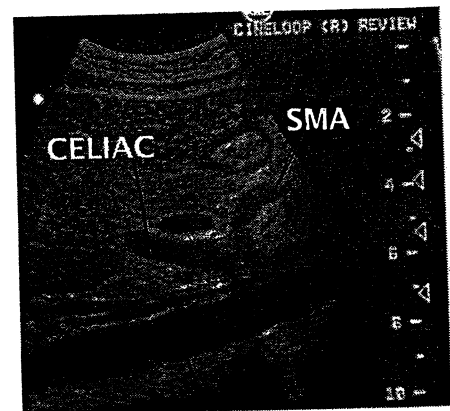
8. MESENTERIC ARTERIES: Technique / Interpretation

Patient history: Patients presenting with a history of dull, achy or crampy abdominal pain 15 - 30 minutes after meals are suspect for **mesenteric ischemia**. The pain, known as mesenteric angina, may be due to a stenosis or occlusion of the SMA, celiac, or IMA (inferior mesenteric artery)

Chronic and acute mesenteric ischemia: difficult to diagnose; require arteriogram.

TECHNIQUE:

- See Abdominal Vessels:
Technique - general remarks.
- Perform exam on FASTING PATIENT.
- Obtain PSV and EDV of the following:
(longitudinal approach)
 - Celiac artery (may use tx approach)
 - Proximal, mid, distal SMA
 - IMA
 - Aorta (check for abnormality)
- Although not routinely done, the food challenge test consists of patient ingesting high caloric liquid (e.g., Ensure). Repeat exam: 20-30 minutes (sooner if symptomatic); hyperemic response begins after about 10 minutes, with maximal response in approximately 30 minutes.
- Post-prandially, the PSV and EDV of the SMA are obtained.



After eating

f. Document:

- Amount of high caloric liquid ingested (usually 1-2 cans),
- Onset, type and duration of symptoms,
- Time began the post-prandial study.

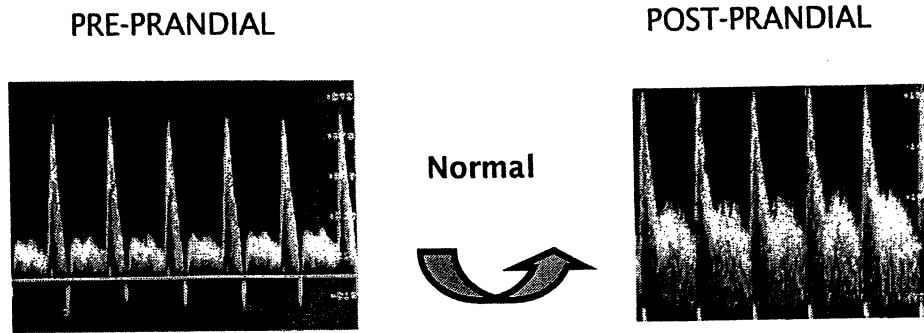
INTERPRETATION:

a.

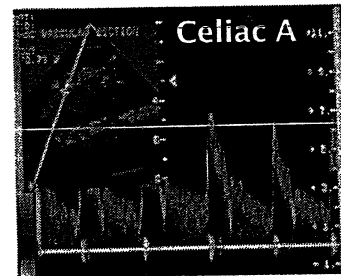
Description	SMA	CELIAC
Pre-prandial (fasting)		
PSV	High	High
EDV	Low	High
Flow reversal	Yes	No
Post-prandial (food challenge)		
PSV	Marked increase	No change
EDV	Marked increase	No change
Loss of flow reversal	Yes	N/A
Velocity criteria		
Normal PSV*	110-177 cm/sec	50-160 cm/sec
Stenosis Criteria*	PSV $\geq$ 275 cm/sec predicts $\geq$ 70% diameter reduction	PSV $\geq$ 200 cm/sec predicts $\geq$ 70% diameter reduction

*Criteria /other info from Gregory Moneta, MD, various papers;
Oregon Health & Science University Hospital, Portland, Oregon

b. SMA Doppler spectral waveform examples:



- c. Celiac Artery branches into the hepatic and splenic arteries. The liver and spleen have fixed metabolic requirements—not likely influenced by the post-prandial state.*



- d. Due to small caliber, difficult to locate IMA off distal aorta. If easily observed, often suggests SMA occlusion.
- e. Observe for proximal stenosis of the SMA and/or celiac A.
- f. In most cases, 2 of 3 mesenteric vessels have to be abnormal to be consistent with chronic mesenteric ischemia.
- g. Extrinsic compression of celiac artery origin by the median arcuate ligament of the diaphragm is frequently seen.
- A reversible celiac artery stenosis occurs during expiration (ligament presses down over artery); rarely the cause of clinical symptoms.
 - The high velocity signals, indicative of stenosis, are substantially improved with deep inspiration.

9. **ORGAN TRANSPLANTS (Allografts)**

(B-mode and Doppler analysis, utilizing a 5 MHz (or 3 MHz) transducer, combine to provide useful information.)

Liver Transplant:

- a. Treats patients with end-stage liver disease.

- b. Pre- and post-operative evaluation includes documenting patency of: portal vein, hepatic veins, IVC and hepatic artery. Observe for tissue abnormalities.
- c. Post-op complications can include thrombosis of vessels such as: portal vein, IVC, and/or hepatic artery leading to hepatic infarction.
- d. Acute rejection will cause liver dysfunction. Many suggest rejection is a cellular process.

Renal Transplant:

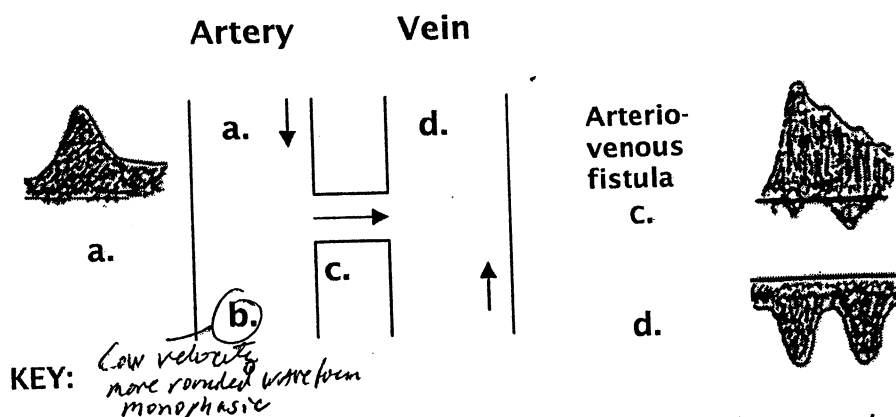
- a. 5 MHz transducer for evaluation.
- b. Both living-related or cadaveric donor kidneys used.
- c. Transplanted RA anastomosed to EIA or IIA; transplanted RV anastomosed to EIV.
- d. Post-op evaluation includes:
 - Doppler evaluation as previously described in this chapter
 - Careful B-mode observation for conditions such as hematoma, and parenchymal echogenicity.
- e. B-mode signs of rejection include: increased renal transplant size, increased cortical echogenicity.
- f. Acute rejection often increases arterial resistance; although other conditions can as well.
- g. Many consider biopsy to be most reliable method for rejection diagnosis; others use the indicator of increased arterial resistance.

Chapter 14

Miscellaneous Conditions / Tests

A. ARTERIOVENOUS FISTULAE (AVF)

1. Abnormal connection between high-pressure arterial system and low-pressure venous system; marked anatomic and hemodynamic changes.
2. Congenital or traumatic.
3. Fistula close to heart, potential for cardiac failure increases
4. Fistula located peripherally, more likely to cause ischemia.
5. AVF may involve proximal and distal arteries/veins as well as collateral arteries/veins. Diameter and length predicts resistance it offers.



- KEY:** *Low velocity more rounded waveform monophasic*
- a. Proximal arterial flow (proximal to the AVF) has greatly increased diastolic flow because the fistula reduces resistance.
 - b. Depending on size of fistula, distal arterial flow may resume its normal triphasic pattern or may be somewhat reduced.
 - c. Flow through the AVF has higher velocities; lower resistant flow
 - d. Venous outflow takes on the flow quality of the fistula's low resistant, more pulsatile flow.

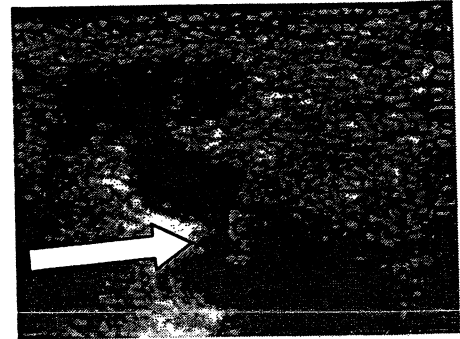
B. COMPRESSION OF PSEUDOANEURYSM

(The following is a very general, simplified summary of this particular procedure)

1. Under guidance of B-mode ultrasound and Doppler color flow image, transducer compression of certain femoral pseudoaneurysms performed.

2. Size and location of pseudoaneurysm, as well as location of "neck" very important to determine.

Key to Success: Whether aneurysm "neck" (communicating channel) between native artery and pseudoaneurysm can be uniformly and completely compressed.



3. Technique:

- Patient signs informed consent.
 - Ideally, anticoagulant therapy discontinued.
 - Dr. present during procedure.
 - Patient somewhat sedated due to discomfort.
 - No standardized protocol at this time, however alternate compression/rest periods occur, with compression lasting about 10-15 min. each
 - Image used to determine if compression maneuver is working, (i.e. thrombosis of aneurysm)
 - Distal monitoring of the great toe arterial pulsations done to make certain there is continued inflow during compression maneuver.
 - Procedure lasts up to 30-60 minutes. Varies widely.
4. Percentage of successful compressions varies. Protocol standardization is important. Increased experience determines overall effectiveness. If procedure unsuccessful, patient usually undergoes surgical correction.

C. **POPLITEAL ARTERY ENTRAPMENT SYNDROME**

1. Popliteal artery possibly compressed by medial head of gastrocnemius muscle (anomalous origin), or fibrous bands.
2. Often found in young men; Repeated trauma to artery may result in development of aneurysm, thrombosis, emboli.
3. Patients present with symptomatic arterial occlusion or intermittent claudication.
4. Flow to the great toe is monitored with an "end point detector" such as PPG. Diminished pulsations considered abnormal.
5. With knee extended and active plantar flexion of the foot (or with passive dorsiflexion of the foot) against resistance, PPG pulsations may diminish or obliterate, which could be suggestive of popliteal artery entrapment.

D. PREOPERATIVE ARTERIAL MAPPING

Epigastric Artery (EA):

1. Deep superior EA is terminal branch of internal mammary artery.
2. Both arteries and perforators, take blood to the rectus abdominis muscle, (long strap muscle) vertically oriented, each side of midline.
3. Rectus abdominis muscle, sub-q fat, arteries, perforators, overlying skin: called the transverse rectus abdominis myocutaneous (TRAM) flap.
4. Reason for mapping: Surgeon wants to use the best arterially supplied muscle section for TRAM flap for autogenous breast reconstruction.

Internal Mammary Artery (Internal Thoracic Artery):

1. Arises off arch of subclavian artery, descending on posterior side of cartilage of upper 6 ribs, about 1 cm from sternum.

2. Reasons for mapping:

Utilized as a recipient site for free flaps in breast reconstruction.

Second important use for this artery is as graft to the left anterior descending (LAD) coronary artery.

Radial Artery Mapping:

1. Determine suitability for use as graft for coronary artery bypass.
2. Technique:
 - Assess patency of palmar arch utilizing modified Allen Test (Chapter 6).
 - If positive, no need to proceed. Removal of radial artery will compromise the hand. If negative, may proceed with exam.
 - Using duplex, (7.5 or 5 MHz) assess brachial and radial arteries
 - Observe for abnormalities such as: focal elevated PSV, abnormal Doppler quality, intimal thickening, aneurysm, calcification.
 - Measure diameter of the radial artery (proximal, mid, distal).
3. Limitations may include: wall calcification

E. PREOPERATIVE VEIN MAPPING

1. Indications:

- Determine suitability for use as extremity or coronary bypass
- Determine suitability of veins for use in dialysis access/graft

2. Technique – LE:

- Transducer: 7.5 MHz or higher
- GSV (sometimes LSV/SSV) mapped (proximal to distal)
- Measurements obtained incrementally along extremity: high, mid, distal thigh; knee, BK, proximal, mid and distal calf, ankle

Technique – UE:

- Cephalic and basilic veins are mapped
- Measurements obtained incrementally along extremity: proximal, biceps, above elbow; elbow, forearm: proximal, mid, distal levels, wrist

3.

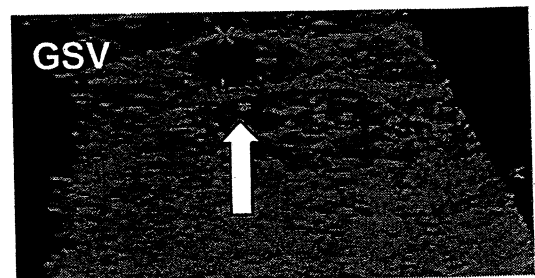
VEIN MAPPING: GENERAL TECHNIQUES

- Proximal tourniquet assists with expansion of the vein size
- Vein wall coaptation (compressibility) is evaluated
- "Outer edge to outer edge" diameter measurements (mm), incrementally along extremity. Vein length (cm) obtained, continuity observed.

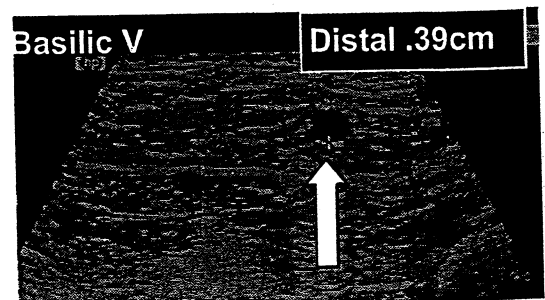
4. Normal findings include:

Compressibility of veins without evidence of thrombosis.

Acceptable diameter measurement criteria varies; However, vein dimensions should be at least 2-3mm; basilic vein is often larger than cephalic vein.



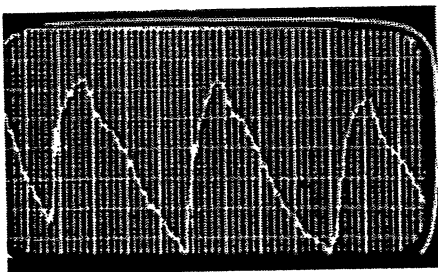
5. Comment made re: presence of vein wall thickening or mural calcifications; findings which could prevent use of vein as bypass graft.



F. THORACIC OUTLET SYNDROME (Arterial component)

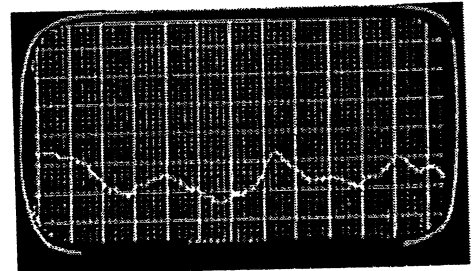
1. Occurs with neurovascular bundle compression by shoulder structures (cervical rib, clavicle, scalene muscle); occurs in certain arm positions.
2. Etiology not well understood; most symptoms due to neurogenic compression of brachial plexus (97%); small percentage due to subclavian vein or artery compression.
3. Symptoms include:
 - Numbness/tingling of arm
 - Pain/aching of shoulder/forearm
 - Exercise/upward positions increase discomfort/symptoms.
 - Note: 25-30% of population have asymptomatic compression.
4. PPG and/or Doppler waveforms utilized to monitor any changes.
 - a. PPG attached to index finger; or CW Doppler on radial artery
 - b. Resting waveforms obtained; patient's arm placed in various positions while arterial pulsations monitored.
 - c. Normal: resting waveforms maintained
 - Abnormal: attenuation or flattening of the waveforms in one or more of the positions.
5. Arm positions (patient sits erect off side of the exam table).
 - a. Resting position - hand in lap
 - b. Arm at 90° angle
 - c. Arm at 180° angle
 - d. Exaggerated military stance
 - e. Adson Maneuver positioning:
 - Same position as d., head turned sharply to right.
 - Same position as d., head turned sharply to left.
 - f. Causative position as described by patient.
6. Example plethysmographic tracings:

NORMAL



Arm position is at
90° with face
away

ABNORMAL



7. Treatment conservative (shoulder exercises) however, surgical treatment done by rib resection with/without scalene splitting.

G. **TRAUMA**

1. Arterial injuries can result from blunt trauma (e.g., long bone fractures which injure vessels) or penetrating trauma (e.g., stab wounds).
2. Clinical presentation may include open wound, hemorrhage and/or hematoma. There may/may not be distal pulses.
3. Intimal tear can also occur with media and adventitia remaining intact. Pulses may remain palpable initially, but events such as thrombus formation, intimal flap or dissection can occur.

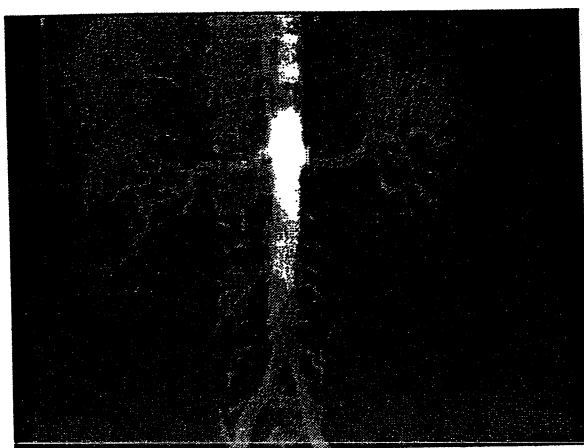
A. ARTERIOGRAPHY

1. Method

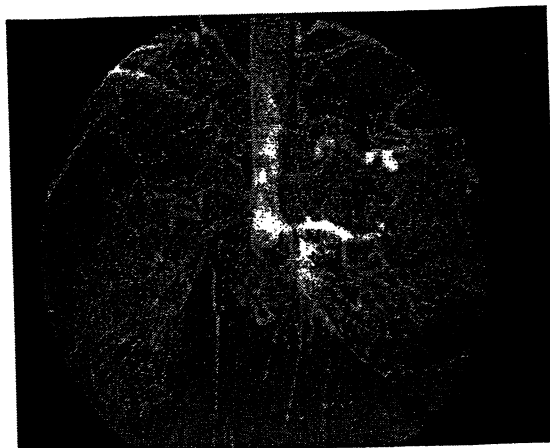
- a. Percutaneous puncture of superficial artery: insertion of thin catheter.
- b. Most common arteries used: CFA (safest approach), axillary, brachial.
- c. After proper positioning, contrast agent injected into catheter, flows with moving blood; gives picture of lumen.
- d. Rapid film changer technique used to expose the films sequentially as contrast agent moves through vessel.
- e. Using fluoroscopy, digital information obtained / stored for later manipulation / interpretation,
- f. Catheter removed; pressure held on puncture site; patient supine for 6-8 hours. Sandbag placed on top of dressing to help avoid bleeding.

2. Interpretation

- a. Primarily an anatomic study NOT a functional study.
- b. Hemodynamically significant stenosis usually defined as a 50% diameter reduction.
- c. Normal anatomy seen on films as contrast media fills vessel.
- d. Interpretation based on how much (if any) of artery does not fill with blood containing contrast agent. Extent and location of filling defect is determined.
- e. Atherosclerotic plaque appears as irregular or smooth.
- f. Vessel occlusion: no filling seen; collaterals often present with long standing occlusion.
- g. Miscellaneous findings:
 - Aneurysm appears as dilated artery
 - Vasospasm: severe narrowing, usually without occlusion.
 - Fibromuscular dysplasia: multiple arterial stenoses caused by medial hyperplasia, appearing as "string of beads."



NORMAL: Aorta, renal, arteries,
common iliac arteries



ABNORMAL: Aortic occlusion
with collaterals; renal arteries
patent

3. Limitations

- a. May be contraindicated in patients allergic to contrast agent or in kidney failure.
- b. Inaccurate in its hemodynamic assessment because of inability to provide many images in multiple planes in "real time." Two-dimensional view is standard.

4. Complications

- a. Puncture site hematoma
- b. Pseudoaneurysm
- c. Local arterial occlusion
- d. Neurologic complications

B. **MR ANGIOGRAPHY (MRA)**

1. Method

- a. Employs radio frequency energy and a strong magnetic field to produce images in multi planes.
- b. MRI instruments quantitate blood flow and construct images that look like angiograms (MRA). Flowing blood well distinguished from soft tissue without using contrast agents.

2. Limitations

- a. Presence of metallic clips, pacemakers, monitoring equipment.
- b. Can overestimate stenosis due to slow flow or turbulence.
- d. Expensive
- e. Claustrophobia may limit some patients.

3. Interpretation

- a. Useful for AAA, dissections, peripheral artery evaluation.

C. **COMPUTERIZED TOMOGRAPHY (CT)**

1. Method

- a. Employs ionizing radiation to obtain cross sectional images of the aorta and other body structures.
- b. IV contrast allows more discrete evaluation.

2. Limitations

- a. Patient motion and presence of metal surgical clips.
- b. One plane used
- c. Limited application in PAD due to smaller vessels

3. Interpretation

- a. Identifies size of aorta; extent and size of aneurysm.
- b. Helps define relationship of aorta to renal artery origins.

D. TREATMENT—MEDICAL

Lifestyle modification:

1. Stop smoking.
2. Consistent exercise to enhance collateral development.
3. Weight control and a low cholesterol diet may enhance normal endothelial cell metabolism.
4. Protection to prevent injury and/or infection to extremity of interest.

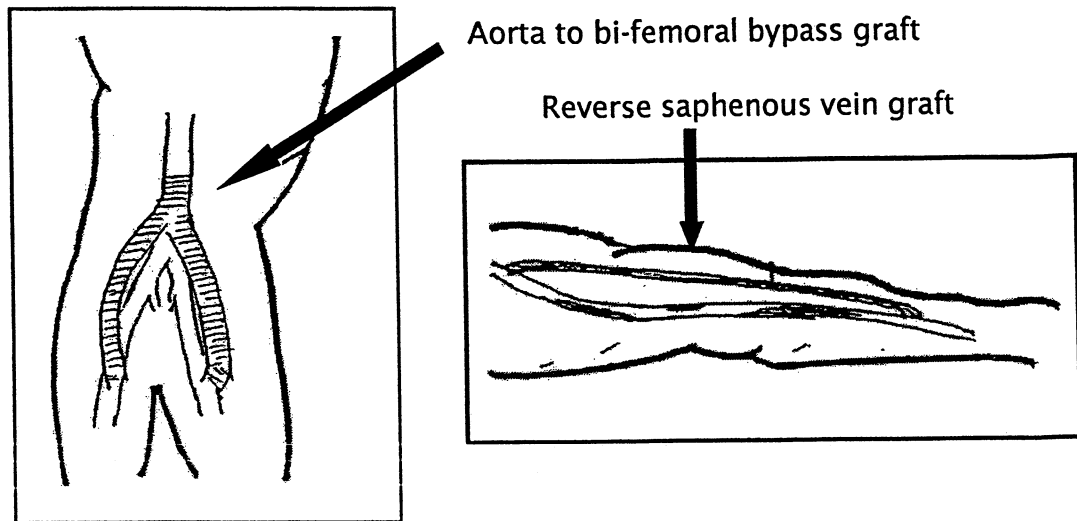
Pharmacologic:

1. Aspirin: an antiplatelet drug that decreases platelet aggregation resulting in decreased thrombotic activity.
2. Medications that help decrease blood viscosity.
3. Antihypertensive drugs may serve to decrease shearing forces against vessel walls.

E. TREATMENT—SURGICAL

Decision to operate depends on extent of disease and primarily the patient's symptoms.

1. Endarterectomy is surgical removal of atherosclerotic material, usually includes portion of intimal lining.
2. In presence of significant stenosis or occlusion, bypass graft provides alternate pathway for blood to travel.
3. Successful bypass grafts include the following components: good inflow, conduit and outflow.
 - a. Common bypass grafts:
 - Aorta to both iliac arteries (also used for AAA)
 - Aorta to bi-femoral (also used for AAA)
 - Femoral to popliteal
 - Femoral to PTA; to ATA, to peroneal



- b. Other bypass grafts include: renal artery, SMA.

C. TREATMENT—ENDOVASCULAR

1. **Angioplasty** (Percutaneous Transluminal Angioplasty [PTLA] used to dilate focal plaque formation in vessel).
 - a. Same technique used for **arteriography**, except a balloon tipped catheter is utilized. Under fluoroscopy:
 - Catheter tip is brought to the region of stenosis.
 - Balloon is slowly inflated, pushing plaque against the walls of vessel, dilating artery lumen.
 - Balloon deflated, catheter removed.
 - b. Can not be done on all vessels nor on all types of lesions. Procedure usually performed in vessels with focal stenosis such as: renal, iliac, femoral, or popliteal artery;

General observations of catheter intervention

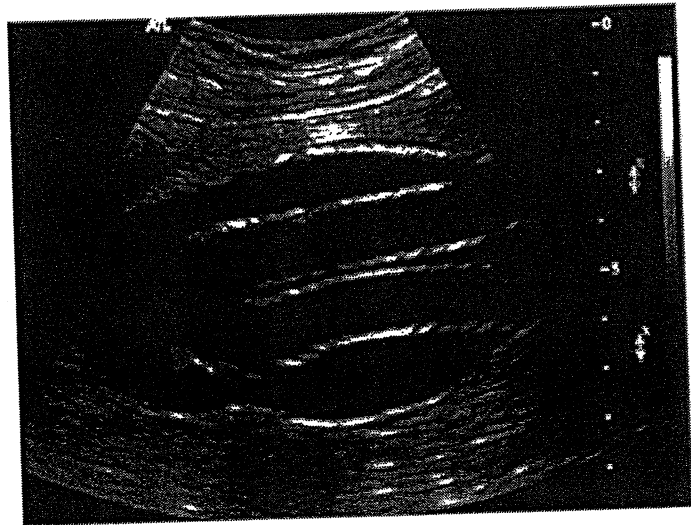
- a. Serious complications occur in less than 5% of cases. 2-3% of PTLA patients have complications requiring surgery or alter hospital stay.
- b. Minor discomfort with the procedure.
- c. Morbidity, mortality, cost of angioplasty are low.

2. Stent / Stent Graft

- a. To maintain intraluminal structure / patency of artery. Acts as a type of "scaffold."
- b. Insertion and deployment technique varies with stent type.
- c. Similar techniques, as used in **arteriography**, are utilized with insertion of stent introducer sheath.
- d. Limitations (depending upon stent site):
 - Abdominal gas
 - Inability of patient to lie flat
 - Complications similar to those for arteriography.
- e. Problems include: restenosis due to intimal hyperplasia, stent migration, graft limb compression, twisting, dislodgment, leaks.
- f. Utilized in vessels such as: aorta, renal, iliac, femoral; repair of AAA with stent grafts.

Aorta
bifurcated
stent graft

Location:
aortic
aneurysm



Can't take multiple pressures in same
extremity at same time
insitu - kept in place - not reversed
WNL - within normal limits
90
thrombosis bilikans -
thrombosis smaller distals
reverse sphenoid - size change